

No.

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**KYSOR INDUSTRIAL CORPORATION'S
COMMENTS ON THE
RI/FS AND PROPOSED PLAN FOR THE
CADILLAC INDUSTRIAL PARK**

EPA Region 5 Records Ctr.



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I. INTRODUCTION.

The U.S. EPA has solicited public comments on the Remedial Investigation/Feasibility Study ("RI/FS") Reports and the Proposed Plan for the Northernair Plating Co. site and the "Kysor of Cadillac, Inc." site. While the Fact Sheet and Proposed Plan imply that the investigations and proposed remedy cover only the Northernair and Kysor sites, there are actually multiple sources in the Cadillac Industrial Park area that are within the scope of the Reports and the proposed remedy. Consequently, these comments refer to the relevant area as the Cadillac Industrial Park area.

Kysor Industrial Corporation ("Kysor") respectfully submits these comments on the RI/FS Reports and Proposed Plan for the Cadillac Industrial Park area. These comments were prepared with technical assistance from Fishbeck, Thompson, Carr & Huber. Fishbeck is an environmental consulting firm with broad experience regarding the issues presented in the RI/FS. Mr. Walter Meinert, a principal and groundwater hydrologist at Fishbeck, has had extensive involvement in hydrogeological investigations in the Cadillac Industrial Park area, including work on the municipal well field.

Kysor requests that U.S. EPA place these comments in the Administrative Record.

II. THE KYSOR SITE IS NOT LISTED ON THE NATIONAL PRIORITIES LIST.

The Proposed Plan and the RI/FS Reports contain statements that the Kysor site is a "Superfund Site." The implicit premise of these statements is that the Kysor site is listed on the National Priorities List ("NPL"). These statements are inaccurate and Kysor believes that this issue should be clarified.

The Kysor site was proposed for placement on the NPL in 1985. Kysor submitted extensive public comments on the NPL proposal, and a copy of the comments is attached as Exhibit A. EPA has not yet acted on the NPL proposal for the Kysor site. Consequently, the Kysor site remains in proposal status and is not listed on the NPL.

In addition, before any action is taken on the Kysor site, the site must be rescored under the revised Hazard Ranking System ("HRS"), which is not yet a final rule. The HRS scoring for the Kysor site was performed under the original version of the HRS. After the Kysor site was proposed for the NPL, the Superfund Amendments and Reauthorization Act was passed, requiring EPA to revise the HRS and implement the revised HRS by October 17, 1988, (CERCLA § 105(c)). After that date, the revised HRS must be applied to any site to be newly listed on the NPL. Since the Kysor site was not placed on the NPL prior to the deadline for the revised HRS, the Kysor site must be rescored before final action is taken on the listing of the site.

In summary, the Kysor site is not listed on the NPL, nor is final action on such listing appropriate until after the site is rescored under the revised HRS.

III. COMMENTS ON THE REMEDIAL INVESTIGATION

A. The Performance Of A Portion Of The RI Work Was Duplicative And Not Cost Effective.

Prior to the EPA's involvement at the Kysor site, Kysor had voluntarily contracted with Fishbeck, Thompson, Carr & Huber to perform an extensive hydrogeological investigation in the area surrounding the Kysor site. Fishbeck performed extensive investigative work, including the installation of twenty-four monitoring wells during the course of a two phase investigation conducted over a three year period. Kysor submitted a report of its findings to MDNR, who later made these findings available to EPA.

Despite Kysor's extensive investigation of the hydrogeology of the area and of the nature and extent of the contamination, and despite all of the findings of this investigation being made fully available to both MDNR and EPA, the agencies nonetheless conducted a second hydrogeological investigation of much of the same area. This investigation was conducted despite Kysor's objections regarding the duplicative nature of the investigation in the area previously studied by Kysor.

The findings in the RI Report on several fundamental matters are essentially identical to those made by Fishbeck for

Kysor. E. C. Jordan's findings with respect to the number of aquifers, the location of the aquifers and aquitards, and the general hydrogeology and groundwater flow direction were the same conclusions previously reached by Fishbeck and Kysor.

In light of the duplicative nature of much of the RI investigation and its findings, the work was unnecessary and not cost effective.

B. The Baseline Risk Assessment Uses Erroneous Assumptions In The Calculation Of Exposure Risk.

1. The Exposure Assessment's Models Are Flawed.

The models used by the EPA in the exposure assessment ("EA") to calculate the baseline risk are much too simplistic to adequately characterize the distribution of contaminants in the shallow aquifer plume at the site. The Agency uses the vertical and horizontal spread ("VHS") model to determine the distribution of groundwater contamination, and the organic leaching model ("OLM") to determine the potential impact of soil contaminants on the groundwater.

Both of these models fail to consider numerous environmental factors which would have a significant impact on the calculations. Neither the OLM nor the VHS model adequately account for biological degradation (known to occur for both chlorinated ethenes and aromatic compounds), chemical degradation, adsorption,

or retardation by soils. The models also fail to account for basic hydrogeological phenomena such as density flow and aquifer recharge, and do not adequately account for dispersion, all of which will undoubtedly result in dilution of the contamination.

The simplicity of the models biases the EA, resulting in overestimation of the risk presented by the contamination. What are characterized as "average" contaminant concentrations in the EA are actually, therefore, upper bound estimates. Realistic modeling would result in lower contaminant concentrations and attendant risks posed by them. Use of a 3-dimensional, coupled-flow-energy-solute-transport model would correct the deficiencies of the OLM and VHS models, and would achieve more realistic results.

2. The Exposure Assessment Uses Implausible Duration Assumptions.

In addition to the use of erroneous contaminant levels, the EA makes implausible and inconsistent assumptions in arriving at a "worst case exposure assessment." On page 7-33, the RI states that a "worst case exposure [for adults] is assessed by using the maximum concentration and assuming that water is ingested for a lifetime" (emphasis added). EPA assumes a lifetime to be 70 years. However, in the same paragraph the RI states that the contaminated plume will impact a given receptor well for only approximately 15 years. There is no rational basis to assume a 70-year period of exposure to contamination which will only impact

the receptor well for 15 years. In effect, the EA bases its analysis on an estimate of the duration of exposure that is 55 years longer than the period supported by the RI.

This error, combined with the excessive contaminant estimates, significantly overstates the baseline risk. The modeling should be revised to more accurately estimate the risk presented by the site.

C. The Remedial Investigation Report Uses Inconsistent And Misleading Designations For The Sources Associated With Four Star And Joynts.

In several areas of the document, the RI Report describes the sources of contamination that were identified during the Remedial Investigation. The RI Report identifies several sources, including the Four Star facility and the former Joynts facility. The former Joynts facility is now the Four Winns facility on Sixth Avenue. Kysor's reference to this facility shall hereinafter be "Joynts (Four Winns Sixth Avenue)."

The RI Report's discussion of the Four Star and the Joynts (Four Winns Sixth Avenue) sources is very confusing and misleading. For example, on pages E-4 and 5-16, the RI Report describes these sources as "Four Star/Joynts." On pages E-5, 1-4, 5-23 and 6-4, the Report refers to these sources as "Four Winns/Joynts" or "Joynts/Four Winns (Sixth Avenue)." These designations confuse the actual facts. The Four Star facility and the Joynts (Four Winns Sixth Avenue) facility have been identified as

separate sources of contamination. The RI should, therefore, refer to the sources separately, rather than confusing the issue by combining the names of the facilities. The RI Report also creates confusion by alternately referring to the "former Joynts facility" as the "Joynts" facility and the "Four Winns" facility. A consistent designation should be used.

For the sake of clarity, the descriptions of these sources should not combine the Four Star facility with the Joynts (Four Winns Sixth Avenue) facility. Instead, the RI and FS Reports should be corrected to list the Four Star and the Joynts (Four Winns Sixth Avenue) facilities as separate sources.

IV. COMMENTS ON THE FEASIBILITY STUDY.

A. The ARAR's Analysis Is Based On The Erroneous Assumption That The Affected Aquifers Are Actual Or Potential Drinking Water Sources.

The ARAR's* analysis in the RI/FS Reports assumes that the affected aquifers are or may be used for drinking water. The reports make this assumption without any detailed analysis of the actual facts regarding the areal limitations of the contamination or the nature of groundwater uses in the affected area.

The relevant facts in this matter indicate that the ARAR's analysis should consider that the affected groundwater is not presently used for drinking water and that future use could be

*A table of terms is attached as Exhibit B.

controlled through institutional controls. The affected groundwater underlies the Cadillac Industrial Park and the North Park subdivision. As recognized in the RI/FS Reports, there are no drinking water wells in use in the affected areas. In fact, municipal water is available throughout the affected area. The RI Report acknowledges that the municipal water supply is not threatened by any of the contaminant plumes other than the Rex-Air plume, since the other plumes are downgradient from the well field. (Remedial Investigation Report, p. E-3.) In addition, the well field is protected by an extensive clay layer overlying the area of the well field. See Exhibit C.

The future use of groundwater in affected areas could be regulated through the use of institutional controls. The county or the state could impose restrictions on the use of the affected areas for drinking water. In light of the availability of municipal water in the area, the actual practical impact of the controls would be minimal.

In light of these circumstances, Kysor believes that EPA should reconsider the assumptions underlying the ARAR's analysis. In particular, the assumption that the affected aquifer will be used for drinking water is not necessarily correct. The actual potential for use as drinking water is minimal and could be eliminated through institutional controls. EPA should, therefore, consider this approach and adjust the ARARs accordingly. Under this more realistic analysis and approach, the scope and cost of the

remedy could be reduced and still be protective of human health and the environment, as required by CERCLA.

B. The Feasibility Study Report Erroneously Applies Certain State Rules And Policies As ARARs.

Even if the affected aquifers are considered drinking water sources, the ARAR's analysis in the RI/FS Reports is incorrect in its application of certain state rules and policies. The standards for the application of state requirements as ARARs are set forth in Section 121 of CERCLA. That section provides that only "promulgated" state standards may qualify as ARARs. As the FS correctly states in Section 2.2: "Section 121 of CERCLA, amended by SARA, requires that hazardous waste remedial actions comply with federal ARARs and state ARARs that are more stringent, legally enforceable, and consistently enforced statewide." Similarly, the Interim Guidance on Compliance With Applicable or Relevant and Appropriate Requirements, 52 Fed. Reg. 32496 (August 27, 1987) ("Interim Guidance"), provides that: "State advisories, guidance, or other nonbinding policies, as well as standards that are not of general application, cannot be treated as requirements under CERCLA."

While the FS Report is vague in its development and application of ARARs, it is apparent that the FS erroneously uses Michigan rules and policies to establish groundwater cleanup standards. On page 2-3, the FS Report relies on the Part 22 Rules of the Michigan Water Resources Act ("Act 245") as the basis for the

adoption of MDNR's "nondegradation" policy. On page 2-6, the FS Report adopts the MDNR's "10⁻⁶" policy. These rules and policies are then applied in the cleanup levels analysis.

The FS Report, however, provides no statutory or regulatory justification for the application of these rules and policies. In fact, there is no meaningful discussion in the Report supporting the application of the rules and policies as cleanup standards. For example, the only justification for the use of the 10⁻⁶ policy is a parenthetical note at the bottom of page 2-6 that this approach is "in accordance with MNDR policy." The FS Report makes no other reference to the statutory or regulatory source of the 10⁻⁶ policy. Further, the Report provides no evidence that the rules or policies were promulgated as cleanup standards, or that they have been consistently applied as such.

1. Michigan Has No Groundwater Cleanup Standards.

The FS Report's failure to justify the use of the Michigan rules and policies as ARARs reflects the lack of actual authority for such use. The MDNR itself recognizes that Michigan law does not presently contain promulgated groundwater cleanup standards. In 1987, MDNR's Director formed a Remedial Action Standards Policy Advisory Committee to investigate the issue of cleanup standards. The Committee recommended that MDNR develop such standards. A 1988 memorandum from the Chief of MDNR's Environmental Response Division addressed the Committee's proposal:

The need for State cleanup standards is also an outgrowth of federal policy. State standards which are formally established and consistently applied will be considered in the remedy selection process for sites addressed through Federal Superfund Programs. The U.S. Environmental Protection Agency is required by Superfund to meet State standards only if they are formally promulgated (emphasis added).

Memorandum from Mr. Guenther, at p. 2 (February 22, 1988).

At the time of the Guenther memorandum, the Part 22 rules and the 10^{-6} policy had been in effect for several years and were obviously not considered to constitute cleanup standards nor provide guidance for that purpose.

Cleanup standards have not been promulgated in Michigan since the Guenther memorandum. The practical result is that there are no promulgated groundwater cleanup standards in Michigan, and it is erroneous to apply the Michigan rules and policies in the cleanup standards analysis.

2. Part 22 Rules Set Standards To Review Applications For New Groundwater Discharge Permits, And Are Not Appropriate For Use As Cleanup Standards.

The Part 22 Rules were intended and should be applied only as standards for assessing discharge permit applications, not as cleanup standards. In "The Groundwater Discharge Permit Process" (March, 1983), at p. 7, MDNR states: "Part 22 Rules (Groundwater Quality) of the General Rules of the Water Resources Commission provide the basis for completing a hydrogeological study" required to apply for a groundwater discharge permit. In

addition, in a February 28, 1989, cover letter to proposed revisions to the Part 22 Rules, MDNR stated that the revisions would: ". . . significantly modify the permitting requirements for several classes of dischargers . . ." (emphasis added). Moreover, the proposed guidelines for Rule 323.2205 [Part 22 rule] state that the guidelines are to be used ". . . in the development of recommendations to the Water Resources Commission on ADLs for discharges of substances to the groundwaters of the State applicable to groundwater discharge permits" (emphasis added).

Consequently, it is clear that these permit standards were not promulgated or intended as cleanup standards.

C. The Feasibility Study Errs In The Calculation Of Residual Risk.

As discussed above, in Section III.B., the Agency's flawed modeling assumptions and failure to consider numerous environmental factors results in overestimation of the baseline risk at the site. These same errors also distort the total risk calculations in Section 2 of the FS Report, resulting in unnecessary modifications of the final groundwater Target Cleanup Levels ("TCLs").

In Table 2-3, the FS Report calculates the total residual risk associated with exposure for 70 years at various groundwater TCLs. As explained in Section III.B.2, the assumption of 70 years of exposure grossly overestimates the actual duration of exposure.

Furthermore, the residual risk calculations assume that the level of contamination will remain at the TCLs for the entire 70-year period. This assumption fails to account for acknowledged environmental factors that will cause the contamination levels to decrease over time. Those factors are natural recharge and dispersion, biological and chemical degradation, adsorption, and volatilization.

The Superfund Public Health Evaluation Manual ("SPHEM") requires that the Agency take these factors into account when accounting for environmental fate and transport. SPHEM states that:

Because the concentration of contaminants changes as substances move from release sources to exposure points, environmental fate and transport must be assessed to project allowable releases. Each exposure pathway will have an identified medium of interest through which the contaminant travels, such as chemicals released to the subsurface that move through groundwater to a well.

SPHEM, p. 95, 102.

In summary, the FS Report errs in calculating the residual risk associated with the TCLs. As a result, the FS Report makes unnecessary and erroneous modifications to the TCLs. The residual risk calculations should be revised to reflect a realistic exposure duration and contaminant concentrations that decrease over time.

D. The Feasibility Study Errs In Setting The Final Groundwater Target Cleanup Levels.

As discussed in Section IV.A., Kysor believes that EPA should reconsider the fundamental assumptions underlying the ARAR's analysis and adjust the ARARs to reflect that the affected aquifers are not drinking water sources. As a secondary position, Kysor believes that the ARARs should be adjusted as discussed below if the aquifers are considered potential drinking water sources.

1. Toluene.

In Table 2-1, the FS Report lists the ARARs for each of the compounds found in the groundwater at the Cadillac site. For the chemical toluene, the chart lists an AWQC of 14.3 mg/l, and a proposed MCLG of 2.0 mg/l. The table does not list an MCL for toluene, despite the fact that an MCL of 2.0 mg/l has been proposed by the EPA. 54 Fed. Reg. 22093 (May 22, 1989), Table 5. The target cleanup level for toluene established in the FS is 0.04 mg/l, based on the proposed Secondary MCL ("SMCL") for toluene.

The FS gives no justification or explanation for not applying either the proposed MCLG or proposed MCL for toluene. The only justification for applying the SMCL appears on page 2-3 of the FS Report, which states that: "MDNR requires the use of Secondary MCLs when available in order to reflect the 'usable' standard of Michigan's Act 245 and Part 22 Rules."

The use of the SMCL and the establishment of a target level below the proposed MCL of 2.0 mg/l must be rejected for at least two reasons: first, SMCLs are not enforceable standards; and second, the proposed MCL for toluene is fully protective of the public health and establishing a more stringent standard is not cost effective, as required by CERCLA § 121.

(a) SMCLs Are Not Proper ARARs Because They Are Not Enforceable.

As noted above, the FS Report relies on MDNR requirements as the basis for applying the SMCL. Only those state standards which are legally enforceable and consistently enforced can be treated as ARARs under CERCLA. SMCLs are neither legally enforceable nor consistently enforced in the State of Michigan. SMCLs have never been officially promulgated in Michigan and are therefore not enforceable by the State of Michigan. Neither may EPA enforce the SMCLs. "National secondary drinking water regulations are federally nonenforceable and set limits for contaminants in drinking water which may effect the aesthetic qualities and the public's acceptance of drinking water (e.g., taste and odor)." See U.S. EPA Fact Sheet, Drinking Water Regulations under the Safe Drinking Water Act, June, 1989, at 34; CERCLA Compliance With Other Laws Manual ("Compliance Manual"), August 8, 1988, at page 4-8.

In addition, the purported basis for the use of the SMCL was the Michigan Part 22 Rules. As discussed above, the Part 22

Rules are not applicable cleanup standards, and it is erroneous to rely on those Rules to establish such standards.

(b) The Use Of SMCLs Is Unnecessary To Protect Human Health And Violates The Cost Effectiveness Requirement Of CERCLA.

CERCLA § 121(b) makes it clear that two of the operative criteria for selecting remedial actions are protection of human health and cost effectiveness:

The President shall select a remedial action that is protective of human health and the environment, that is cost effective, and that utilizes permanent solutions and alternative treatment technologies or resource recovery technologies to the maximum extent practicable.

SMCLs are not designed to reduce public health risks. Rather, they are simply intended to improve the aesthetic quality of drinking water.

MCLs, in contrast, are considered fully protective of the public health, which is the applicable criterion. See Interim Guidance, Compliance Manual at page 4-3. Furthermore, the proposed MCL for toluene is equal to the MCLG proposed for toluene by the EPA. According to the Interim Guidance, "MCLGs are based entirely on health considerations and do not take cost or feasibility into account. Moreover, as health goals MCLGs are set at levels where no known or anticipated health effects may occur, including an adequate margin of safety." (Emphasis added). Since the proposed MCL is equal to the MCLG, there is no conceivable health basis for not using the proposed MCL.

Establishing a cleanup level for toluene which is more stringent than both the proposed MCLG and MCL for the chemical would only result in additional cleanup work and costs that are not necessary to protect human health. The unnecessary cleanup and related costs would violate CERCLA's cost effectiveness requirement.

2. Tetrachloroethylene ("PCE").

The FS Report establishes the final target cleanup level for tetrachloroethylene (or "PCE") at .001 mg/l. The procedure used to arrive at this figure is somewhat vague and confusing. The FS states that the only ARAR for PCE is an AWQC of 0.0008 mg/l. This statement fails to recognize that the EPA has proposed an MCL for PCE of 0.005 mg/l. Since the AWQC level is below currently available detection limits, a second figure, based on a cancer risk of 10^{-6} , was calculated to be 0.001 mg/l.

As established in Section IV.B., there is no justification for the selection of a 10^{-6} risk level as the proper criteria for a cleanup standard, beyond a mere parenthetical reference to "MDNR policy." Further, the FS fails to consider the fact that a level of 0.001 mg/l is also below the practical quantitation level ("PQL"), as established by the EPA. The PQL for a given contaminant is defined by the EPA as "the lowest concentration that can be reliably measured within specified limits of precision and accuracy during routine laboratory operating conditions." 50 Fed.

Reg. at 22100. The PQL for PCE is 0.005 mg/l. In setting the cleanup target level at .001 mg/l, the FS Report imposes an unattainable requirement that is unnecessary for the protection of human health and the environment.

The proper approach is to adopt the proposed MCL as the cleanup target level for PCE. PCE has a proposed MCL of .005 mg/l. 54 Fed. Reg. 22093, May 22, 1989, Table 5. Under the SDWA, MCLs must be set "as close to the maximum contaminant level goal as is feasible." SDWA § 1412(b)(4) The term "feasible" means "feasible with the use of the best technology, treatment techniques, and other means, which the Administrator finds, after examination for efficacy under field conditions and not solely under laboratory conditions, are available (taking cost into consideration)." SDWA § 1412(b)(5). Because of the stringency of the MCLs, the Interim Guidance recognizes that MCLs are fully protective of human health and should be applied when available. In accordance with the Interim Guidance, the proposed MCL for PCE should be applied in this case.

3. 1,1-Dichloroethylene ("1,1-DCE").

As has been discussed above, MCLs are fully protective of the public health and should be used to establish the cleanup level whenever available. This is especially true when an MCL is equal to a compound's MCLG, a figure set without taking cost or feasibility into account. The MCL for 1,1-DCE is equal to its MCLG of 0.007 mg/l. Despite this fact, the FS Report establishes

a cleanup level for 1,1-DCE of 0.005 mg/l. The apparent basis for the failure to use the MCL for 1,1-DCE was the calculation of the additive risk level. That calculation, however, was based on erroneously conservative assumptions as discussed in Section III.B. In addition, EPA has recognized that the additive risk calculation and the 10^{-4} - 10^{-7} risk range are inappropriate as strict requirements when the resulting cleanup levels are infeasible. For example, the total risk level allowed at the U.S. Aviox site in Michigan was 1.18×10^{-4} . See Summary of Remedial Alternative Selection, U.S. Aviox Site, Exhibit D. Consequently, in accordance with the Interim Guidance and the need to set achievable and cost effective standards, the MCL for 1,1-DCE should be applied in this case.

4. Trans-1,2-Dichloroethylene ("T-1,2-DCE").

The cleanup level for T-1,2-DCE established in the FS is 0.07 mg/l, and is erroneously stated to be equal to the proposed MCLG for the compound. In fact, the EPA has re-proposed the MCLG for T-1,2-DCE at 0.1 mg/l. 54 Fed. Reg. 22070, Table 4, with discussion at page 22083-84. The EPA has also proposed an MCL for T-1,2-DCE of 0.1 mg/l. 54 Fed. Reg. 22093, Table 5.

Neither the FS discussion nor Table 2-1 include either the re-proposed MCLG or the proposed MCL for T-1,2-DCE. Since the FS does not state a justification for establishing a cleanup level for T-1,2-DCE below its proposed MCLG or MCL, these comments cannot evaluate the Agency's procedure or reasoning. However, as

has been discussed above, both the MCLG and MCL are fully protective of the public health, and should be used to establish the target cleanup level for T-1,2-DCE.

E. The Feasibility Study Errs In Setting The Final Soil Target Cleanup Levels.

As part of the soils cleanup levels analysis, the FS Report applies the Organic Leaching Model ("OLM"). As discussed in Section III.B., the OLM does not accurately characterize the distribution of contaminants at the site. The model fails to account for biological degradation, chemical degradation, adsorption, volatilization, dispersion, and density flow. For all of these reasons, as stated in Section III.B., the OLM overestimates the risk posed by given levels of soil contamination, resulting in TCLs being set at levels lower than necessary to protect the public health and environment.

Furthermore, the procedure for determining soil TCLs uses the groundwater TCLs and then back-calculates to arrive at soil TCLs. In effect, the groundwater TCLs are the beginning point in the analysis. If the groundwater TCLs are incorrect, the calculation of the soils TCLs is flawed. As established in Section IV.D., the FS Report erred in the selection of the groundwater TCLs. As a result, the soils TCLs must be recalculated based on the correct groundwater TCLs.

In the soils TCLs analysis, the FS Report also erroneously states that the proposed MCLG for xylene is 0.44 mg/l. In

fact, the EPA has changed the proposed MCLG for xylene to 10.0 mg/l, and has proposed an MCL for xylene at the same level of 10.0 mg/l. The EPA stated:

EPA proposed an MCLG of 0.44 mg/l for xylenes in the November 1985 proposal. However, since that time new data has become available which change EPA's conclusions about xylene's toxicity . . . resulting in a re-proposed MCLG of 10 mg/l for xylenes.

54 Fed. Reg. 22092, May 22, 1989.

The proper groundwater TCL for xylene is equal to its proposed MCL of 10 mg/l, and this should be reflected in the calculation of the soils TCLs.

F. The Feasibility Study Report Overestimates The Requirements For The Groundwater Treatment System Because Of The Use Of Unreasonably High Estimates For The Purge Water Contaminant Concentrations.

The FS Report's analysis of the groundwater treatment systems is flawed because it is based on a completely unrealistic analysis of the estimated contaminant concentrations in the purge water. The practical consequences of this error are significant. The result is that the FS Report calls for more treatment capacity than necessary for an effective cleanup. If the actual design is based on this analysis, the cleanup system will have excess capacity and involve unnecessary capital and operating costs.

The underlying problem is that the FS Report uses a faulty analysis to estimate the contaminant concentrations in the purge water. While the Feasibility Study does not directly

present estimated contaminant concentrations in the purged groundwater, E. C. Jordan personnel have confirmed in telephone conversations that they used the contaminant concentrations listed in Table 2-11. That table lists the maximum contaminant concentrations detected for each compound over the entire site during the course of the investigation. This approach does not account for the vertical distribution of contaminants, the dilution that occurs as the contaminants are drawn in radially from all sides of each purge well, the degradation that occurs over time, or the fact that contaminant concentrations in some of the plumes are lower than others.

The impact of these factors is significant, and the contaminant concentrations in the purged groundwater will actually be much lower than those indicated in Table 2-11. For this reason, the FS Report overstates the requirements for the groundwater treatment system.

The following discussion further illustrates why the assumed purge water contaminant concentrations are too high. Table 2-1 lists existing maximum contaminant concentrations for each of the four major plumes. Table 2-11 lists the maximum concentration detected for each contaminant at the entire site. Based on the samples collected during the remedial investigation and other data, a more accurate estimate of the influent contaminant concentrations could be obtained by using the information in Table 2-1 and Table 7-4 to estimate purge water contaminant con-

centrations by calculating flow-weighted-average concentrations. Using the information from Table 7-4, the following pumping rates can be attributed to each of the four major plumes:

- (a) Northernnaire - 32 gpm
- (b) Rex-Air - 135 gpm
- (c) Shallow VOC - 330 gpm
- (d) Intermediate VOC - 387 gpm

The table below shows the flow-weighted-average concentration for each of the contaminants listed in Table 2-1 and compares these with the concentrations from Table 2-11.

ESTIMATED PURGE WATER CONTAMINANT CONCENTRATIONS (mg/l)

<u>Contaminant</u>	<u>Existing Maximum Concentration from Table 2-1</u>				<u>Flow-Weighted Average Concentration</u>	<u>Table 2-11 Maximum Concentration</u>
	<u>Rex-Air Plume</u>	<u>Shallow VOC Plume</u>	<u>Intermediate VOC Plume</u>	<u>Northernnaire Plume</u>		
1,1,1-TCA	0.017	32.000	1.200	0.440	12.490	32.000
Trans-1,2-DCE	0.048	0.000	4.600	0.040	2.023	4.600
1,1-DCE	0.000	1.000	0.099	0.026	0.418	1.000
1,2-DCA	0.000	2.000	0.000	0.000	0.747	2.000
Meth. Chl.	0.078	0.690	0.240	0.000	0.375	0.690
PCE	0.000	0.720	0.170	0.003	0.343	0.720
TCE	153.000	78.000	7.500	2.500	55.857	153.000
Chromium	0.000	0.000	0.000	0.773	0.028	0.773
Xylene	0.000	1.700	0.000	0.000	0.635	N/A
Toluene	0.042	3.200	0.001	0.000	1.201	3.200

The flow-weighted-average concentrations are significantly lower than the maximum concentrations reported in Table 2-11, which were used to evaluate potential treatment systems and estimate costs. This difference will have a significant effect on the size of the required treatment system and on the

capital costs and annual operating and maintenance costs of that system.

This rather simplistic analysis is used here only to demonstrate that more accurate estimates of contaminant concentrations are possible. This analysis does not account for the vertical distribution of contaminants or the dilution effects when the purge wells are operated, which would further reduce the estimates. Prior to designing the groundwater treatment system, groundwater modeling should be performed to more accurately predict the contaminant concentrations in the purged groundwater. The use of more accurate estimates is critical to ensure that the system is optimally designed for the actual contamination and, therefore, cost effective. The modeling issue is discussed further in Section VI.

G. The Feasibility Study Underestimates The Time Required For Implementation Of The Groundwater Remedy.

The estimated times required to implement most of the groundwater cleanup alternatives discussed in the FS Report are unrealistic. In particular, the preferred groundwater treatment alternative, Alternative 3-A, is estimated to require 3 to 6 months to install. The actual time to complete all of the necessary work would be significantly longer. The time required to complete groundwater modeling efforts and design the treatment system would be 12 to 16 months. Following the design, bids would be taken for construction of the treatment systems, requiring

another 2 months. Construction of the entire system, including extraction wells, influent piping, discharge piping, treatment building, and treatment equipment would probably require approximately 12 to 16 months. A realistic estimate for the entire project proposed by EPA is approximately 2 to 2 1/2 years. This construction schedule is consistent with the timetable anticipated by the cogeneration project summarized in Section VII.

H. The Feasibility Study Should Evaluate Thermal Oxidation As An Alternative For Treatment Of Off-Gas From The Air Stripper.

The FS Report investigated only one treatment alternative, Vapor Phase Carbon Adsorption, as a treatment method for the off-gas from the air-stripping tower. The FS Report failed to evaluate the feasibility of thermal oxidation. Thermal oxidation or catalytic combustion are methods that have been successfully utilized to treat air stripper off-gas containing VOCs. The VOC contaminants present in the Cadillac area groundwater do not present any unusual complications that would preclude the use of thermal oxidation. The evaluation of this alternative is particularly appropriate in this matter, since the proposed cogeneration project discussed in Section VII would provide cost-effective thermal oxidation for the off-gas. Kysor believes, therefore, that this alternative should be evaluated and included in the Record of Decision for this site.

I. The Feasibility Study Should Evaluate The Accelerated Purge Rate That May Be Associated With The Cogeneration Project.

The proposed cogeneration project, discussed in Section VII, is expected to involve an overall purge rate that is significantly higher than the rate associated with Groundwater Alternative 3A. The purge rate for the cogeneration project may be as high as 5000 gpm, compared to the 940 gpm for Alternative 3A.

The increased purge rate would have several effects. It would significantly accelerate the completion of the cleanup. In addition, the discharge to the Clam River would be increased. For example, if the purge rate is 5000 gpm, the discharge to the Clean River would be 4400 gpm, and 600 gpm would be evaporated from the cooling towers.

Kysor is unaware of any problems associated with increased purge rates. This approach, however, should be evaluated in the Feasibility Study and accounted for in the Record of Decision.

J. The Feasibility Study Report Uses Misleading Terminology To Identify The Extraction Well Systems.

The Feasibility Study Report's terminology describing some of the extraction well systems is inappropriately misleading. The extraction well systems designed to purge the Shallow VOC plume and the Intermediate VOC plume are identified as Kysor Shallow (A), Kysor Shallow (B), and Kysor Intermediate. This implies

that these extraction well systems are removing contamination originating solely from Kysor.

In fact, the RI and FS Reports identify several contaminant sources for both the Intermediate and Shallow VOC plumes. Section 1.2 of the FS Report describes seven possible or known contaminant sources: Northernnaire, Kysor, Rex-Air, Four Winns (Frisbie Street), Four Star, Joynts (Four Winns Sixth Avenue), and Ingraham property. Section 1.2 also identifies four major plumes: Northernnaire plume, Shallow VOC plume, Intermediate VOC plume, and Rex-Air plume.

A specific analysis of the RI/FS Reports' discussion of the sources and plumes demonstrates that there are multiple sources for each plume. On page 1-3 of the Feasibility Study, Kysor and Ingraham are referred to as sources of the contamination in the Intermediate VOC plume. On page 7-7, the Feasibility Study describes chemicals originating from Four Star and Joynts (Four Winns Sixth Avenue) in the Intermediate VOC plume. In addition, the northern most extraction well for this system is downgradient of the Four Winns (Frisbie Street) site. With the aquitard between the shallow and intermediate aquifers ending in this area, it is apparent that VOC contaminants from the Four Winns (Frisbie Street) site are present in the northern portion of the Intermediate VOC plume. Therefore, the extraction well system termed "Kysor Intermediate" will be removing contaminants that originated

from Kysor, Ingraham, Four Star, Joynts (Four Winns Sixth Avenue), and Four Winns (Frisbie Street).

The shallow plume is also a multiple source plume. On page 1-3 of the Feasibility Study, Kysor, Ingraham, Four Star, and Joynts (Four Winns Sixth Avenue) are described as sources of the contaminants in this plume. On page 7-21 of the Feasibility Study, it is indicated that VOC contaminants from Four Star, Joynts (Four Winns Sixth Avenue), and Four Winns (Frisbie Street) are evident in the Shallow VOC plume. Therefore, the extraction well systems termed Kysor Shallow (A) and Kysor Shallow (B) are actually designed to remove contaminants that originated from Kysor, Ingraham, Four Star, Joynts (Four Winns Sixth Avenue), and Four Winns (Frisbie Street).

In light of the multiple sources involved in each of the plumes, it is unfairly misleading to name the extraction well systems as "Kysor" systems. The names should be revised to reflect that multiple sources are involved in each aquifer.

V. COMMENTS ON THE PROPOSED PLAN.

A. The Proposed Plan Is Inaccurate And Misleading.

The Proposed Plan is inaccurate and misleading on three significant issues. First, the Proposed Plan implies that Northernnaire and Kysor are the only sources of contamination in

the Cadillac Industrial Park area, when there are actually numerous other sources. Second, the Proposed Plan states that the recommended remedy is for the Northernnaire and Kysor sites, when the remedy addresses several other sources as well. Third, the Proposed Plan and the Feasibility Study understate the portion of the remedial costs attributable to the hexavalent chromium.

The Proposed Plan identifies two sources of contamination in the Cadillac Industrial Park area: Northernnaire and Kysor. The clear, but inaccurate, implication is that those sites are the only sources of contamination in the area. In fact, as reflected in the RI/FS Reports, there are several other sources of contamination in the area. The other facilities identified in those reports as contaminant sources are: Four Winns (Frisbie Street), Four Star, Joynts (Four Winns Sixth Avenue), and Rex-Air. Since the Proposed Plan is intended to provide a detailed summary to allow an effective evaluation of the proposed remedy, it should provide a complete description of the contaminant sources and their contribution to the problem at issue.

Similarly, the Proposed Plan should accurately describe all of the sources and areas of contamination that would be addressed under the proposed remedy. Instead, the Proposed Plan erroneously states that the proposed remedy is for the Northernnaire and Kysor sites. The proposed remedy would actually address contamination from several other sources, including, Four Winns (Frisbie Street), Four Star, Joynts (Four Winns Sixth Avenue) and

Rex-Air. Consequently, the Proposed Plan should identify each of the sources within the scope of the remedy and the underlying Reports and/or the Proposed Plan should justify the inclusion of those sources.

In addition, the Proposed Plan and the Feasibility Study fail to accurately state all of the costs properly attributable to the hexavalent chromium contamination from the Northernnaire facility. The only costs attributed to that source are the treatment costs. The proper approach is to also attribute a significant portion of the costs for the treatment building and for the installation and operation/maintenance of the extraction and discharge systems.

This approach is particularly appropriate when the nature of the contamination is obviously distinct from the contamination from other sources. In this case, the hexavalent chromium is attributable solely to Northernnaire and none of the other PRPs has any liability for the costs attributable to the hexavalent chromium. Under these circumstances, EPA has an obligation to make a fair and accurate allocation of the costs attributable to the Northernnaire source.

B. The Scope Of The Groundwater Remedy Is Unnecessarily Broad.

As discussed above, the Proposed Plan states that the proposed remedy is for the Northernnaire and Kysor sites. That is obviously inaccurate and unfairly focuses on only two of the

sources in an area-wide problem that EPA proposes to address with an area-wide remedy. The Feasibility Study makes it clear that the remedy also includes Four Winns (Frisbie Street), Four Star, Joynts (Four Winns Sixth Avenue), and Rex-Air, with the possibility of including the plumes in the North Park subdivision.

Some of these sources are included even though the related areas of contamination are entirely distinct from the areas allegedly affected by Northernnaire and Kysor. The RI/FS Reports make it clear that Rex-Air is a separate source that affects an area upgradient from the City of Cadillac well field. None of the Rex-Air contamination has commingled with the plumes attributed to Northernnaire or Kysor. In short, Rex-Air is a clearly distinct source that affects a separate area.

Similarly, the RI/FS Reports attribute the plumes in the North Park area to the Four Star and Joynts (Four Winns Sixth Avenue) facilities. The plumes in this area are considered separate and distinct from the plumes in the Cadillac Industrial Park area.

Since these sources and affected areas are distinct, there is no technical necessity for including these areas in the proposed remedy for the Northernnaire and Kysor sites. In fact, it appears that these areas are not eligible for fund financed remedial action (NCP § 300.68), and should not be included in a proposed CERCLA remedy. Consequently, if these areas are included,

the remedy is unauthorized and unnecessarily broad, and the costs are correspondingly inflated.

C. The Agency Should Reconsider Soil Flushing And Other Technologies As The Soils Remedy For The Kysor Site.

In the FS and Proposed Plan, the EPA evaluates various alternatives for the cleanup of the contaminated soils at the Kysor site. In the Proposed Plan, the EPA's preferred alternative is "In-situ Vacuum Extraction." This alternative has an estimated present-worth cost of \$925,000.

Neither the Proposed Plan nor the FS give any adequate justification for selecting the Vacuum Extraction alternative over "In-situ Soil Flushing," an equally effective alternative at a cost of \$225,000, which is less than one-fourth the cost of Vacuum Extraction.

The Proposed Plan, at page 6, acknowledges that both alternatives provide an equally effective, permanent, long-term remedy. Table 8-8 in the FS presents a summary of the detailed analysis of the Soil Flushing alternative. This table indicates that the Soil Flushing alternative satisfies all of the criteria used to evaluate the alternatives, and "present[s] the lowest risk to on-site workers during installation and implementation." While the purpose of Soil Flushing is to increase the mobility of the groundwater contaminants temporarily, the Proposed Plan provides for an extensive groundwater capture system, which will work in

conjunction with the Soil Flushing procedure, to eliminate this contamination.

The Soil Flushing technique has been successfully implemented at various sites in Michigan including the U.S. Aviex site in Niles, Michigan, and the McGraw Edison site in Albion, Michigan. See Exhibits D and E. EPA has selected this technique at other CERCLA sites, including the Goose Farm site in New Jersey. See Exhibit E. In Section 8.5.3 of the FS, the EPA acknowledges that "In-situ Soil Flushing technology has been proven successful in hydrogeological conditions similar to those at the Kysor and Four Winns sites." Yet, neither the FS nor the Proposed Plan give any justification for choosing an alternative which will cost an estimated \$700,000 more than the proven effective Soil Flushing method. Since the Soil Flushing alternative satisfies the applicable criteria, and is more cost effective, EPA should reconsider that alternative.

In addition, the agency should reconsider an aspect of the vacuum extraction alternative. Kysor understands that there is an additional technology for treating the soil gas from the extraction system. Recent research has demonstrated the effectiveness of using soil composting techniques to treat the soil gas. This technology should be evaluated prior to final remedy selection.

D. The Soils Remedy For The Kysor Site Does Not Require The Use Of Deed Restrictions.

The Proposed Plan would impose deed restrictions on the Kysor site which would "prevent building or excavating soils from the Kysor site to ensure there are no new releases or contamination from soils to the environment." These deed restrictions are completely unnecessary and overburdensome and would not result in any additional protection to human health or the environment.

Deed restrictions may be an appropriate protective measure in those situations where contaminants remain in the soil. For example, in Alternative 1 - No Action, deed restrictions might be necessary to protect humans from exposure to the contaminants. However, the chosen alternative - Vacuum Extraction, and the alternative recommended in these Comments - Soil Flushing, both provide a permanent remedy and would reduce the level of contaminants in the soil to the TCLs selected for the site as being fully protective of the public health and environment.

Neither the Proposed Plan nor the FS provides any justification for imposing deed restrictions on what will be "clean" property after just 2 years. In fact, the FS never even mentions deed restrictions in connection with any of the alternatives except "No Action." Nor does either document justify imposing deed restrictions on the entire site, when the area of soil contamination is very limited. Imposing deed restrictions on the

entire site would unfairly deprive Kysor and future owners of the use of portions of their land that were never contaminated.

According to Section 8.6.8 of the FS Report, after the soil remediation is completed "there would not be any hazardous substances, pollutants, or contaminants remaining on-site" The FS Report concludes, therefore, that not even the typical 5-year site review would be required under SARA Section 121. Deed restrictions are completely inappropriate in these circumstances.

VI. GROUNDWATER MODELING IS NECESSARY PRIOR TO DESIGN AND IMPLEMENTATION OF THE REMEDY.

The complexities involved in this matter require the development and use of a refined groundwater model to ensure the proper design of the cleanup system. Based on the present data, there is insufficient information on several issues critical to the design of the system. For example, the present data and analyses do not properly evaluate the potential cost benefit of multiple purge system scenarios or the potential benefits of accelerated purging. In addition, as discussed in Section IV.F., the FS Report overestimates the VOC concentrations in the purge water, which will result in excess treatment capacity and unnecessary costs due to overdesign of the system.

The use of a proper groundwater model can provide the necessary data and result in the optimal design for the groundwater cleanup system. A groundwater flow/chemical transport model will meet those objectives. The model should be a 3-dimensional

numerical model to allow evaluation of vertical distribution, chemical transport and flow. This model will provide realistic purge water contaminant concentrations and permit rapid evaluation of various purge well/extraction scenarios. In addition, the model will perform ongoing operational monitoring and result in more effective operation.

The benefits of the model are significant. It will ensure that the cleanup system is optimally designed to provide a complete groundwater cleanup in the most cost effective manner. In addition, the model will provide information that will improve the likelihood of a cooperative and joint implementation of the remedy. In particular, the model would provide information that would assist in the allocation of responsibility for the area-wide contamination.

In light of the present inadequacy of the data, which will result in improper design of the cleanup system, groundwater modeling should be conducted prior to the design phase of the project.

VII. THE COGENERATION PROJECT PROPOSED BY CMA PROVIDES THE APPROPRIATE GROUNDWATER REMEDY FOR THE AREA GROUNDWATER CONTAMINATION.

Cogeneration Michigan Associates ("CMA") is in the final stages of the organization of a cogeneration project. The project will be constructed and operated in the Cadillac Industrial Park. The project as proposed will generate electrical power and steam,

and provide an accelerated groundwater cleanup for the area-wide contamination in the Cadillac Industrial Park area.

CMA is a limited partnership that was formed for the ownership and operation of the project. The general partner is Cogeneration Michigan, Inc. ("CMI"). The shareholders of CMI are substantial entities: City of Cadillac, Hagar Industries, Townsend & Bottum, and Kysor Industrial Corporation. In addition, Consumers Power Company has expressed significant interest in an equity participation in the project and involvement in its operation.

CMA initiated the cogeneration project several years ago and experienced early delays associated with administrative approvals and proceedings. CMA, however, has overcome the administrative delays and made substantial progress during the past year towards actual implementation of the project. Earlier this year, the Michigan Public Service Commission ("PSC") issued an opinion that approved several fundamental aspects of the cogeneration project. CMA has been advised that the PSC will grant Act 81 approval for the project by September 1, 1989.

In addition, the basic operational and financial arrangements for the project have been completed or are essentially complete. CMA has entered into a 35-year Power Purchase Agreement with Consumers Power Company. Under the Agreement, Consumers will purchase the electrical power generated by the project. The City of Cadillac will purchase the steam and sell it to industrial

users. In addition, arrangements have been made for Hagar Industries to operate the lumbering and sawmill operations for the project. The financial arrangements have been handled through Robert W. Baird & Co., which is working with Goldman Sachs.

In short, CMA has made substantial progress on critical issues, and the cogeneration project is very nearly ready for implementation. CMA anticipates that it will begin actual construction of the project before the end of 1989.

The cogeneration project deserves special consideration because it provides an opportunity for an innovative solution to an environmental problem. The power and steam generated by the project will provide economic benefits and stimulation for the community and area industry. At the same time, the project will provide an accelerated and cost effective cleanup for the area groundwater contamination. In effect, the project provides the unique combination of economic benefits and an effective environmental cleanup.

The cleanup capabilities of the cogeneration project are consistent with, or better than, the groundwater remedy selected in the Proposed Plan. The project will include a sufficient number of purge wells to capture the plumes in the industrial park area. The exact location of the wells and the purge rates will be determined with the results of a numerical 3-dimensional coupled flow model.

The proposed treatment is also consistent with EPA's proposed remedy. The project will include conventional air stripping equipment, as described in the Proposed Plan. The cogeneration project is different in one respect regarding treatment. The project proposes to treat the off-gas from the air strippers through thermal oxidation in a boiler. As discussed in Section IV.H., this treatment is effective for VOCs, and should be evaluated and included in the Record of Decision.

Another significant difference between the cogeneration cleanup and the agencies' proposed remedy is the rate of cleanup. The cogeneration project may involve a purge rate of up to 5000 gpm, compared to the 940 gpm for Alternative 3A. This difference should result in a significant acceleration of the cleanup, which will reduce operating costs and more quickly achieve the final goal of the agencies' efforts.

In light of the environmental and economic benefits of the cogeneration project, Kysor believes that it offers a uniquely effective solution to the area-wide problem in the Cadillac Industrial Park. In effect, the project offers an opportunity for a cooperative area-wide solution that will benefit the agencies, the community, and area industry, in contrast to the inefficiencies inherent in separate efforts by each of the PRPs and the agencies.

Kysor requests that the agencies consider and evaluate the project, and account for it in the Record of Decision.

VIII. CONCLUSION.

These comments address critical issues that will have an effect on the cleanup levels and the design of the cleanup systems. Kysor requests that U.S. EPA carefully review these issues and take appropriate action prior to entry of the Record of Decision.

Respectfully submitted,

KYSOR INDUSTRIAL CORPORATION

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Vice President - General Counsel

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Exhibit A

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U.S. Environmental Protection Agency
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Re: Fourth Proposed Update to the
National Priorities List,
50 Federal Register 37950,
September 18, 1985

Dear Mr. Wyer:

Kysor Industrial Corporation ("Kysor") respectfully submits these comments on the fourth proposed revision of the National Priorities List ("NPL"). Kysor's comments will address the U.S. Environmental Protection Agency's ("EPA") proposal to add Kysor's Cadillac, Michigan facility to the NPL.

Kysor has two general objections to EPA's proposal:

(1) the Hazard Ranking System ("HRS") score calculated for the Kysor site is too high because of errors in applying the scoring methodology; and (2) the proposal erroneously lists Kysor as the site name when a more general site name should be used since other non-Kysor facilities have contributed to the contamination in the aquifer of concern as defined by EPA.

The remainder of these comments will discuss these objections in detail. Kysor's comments are based on a careful review of the HRS scoring documents, and the governing provisions of CERCLA, the NPL and the HRS.

In reviewing these comments, EPA should note that there are several alleged sources of contamination in the area involved in this HRS scoring. No determinations have been made as to the extent, if any, that any particular source has contributed to the contamination. In particular, it has not been determined that Kysor has contributed to any existing contamination and none of these comments should be interpreted as an admission that Kysor has done so.

I. Erroneous HRS Scoring.

The groundwater route of the migration hazard mode was the sole basis for the proposed Kysor HRS score. The other routes and modes were correctly given scores of zero. The score for the groundwater route was overstated because of errors in calculating the "Population Served" component of the Targets score.

Two errors were committed in calculating the "Population Served" score. First, the target area of the aquifer of concern was defined too broadly causing an exaggeration of the actual population at risk. Second, regardless of the proper definition of the aquifer of concern, improper methodology was used to count the population served by the aquifer.

A. Definition of Target Area of Aquifer of Concern.

There is no dispute that the aquifer of concern was correctly defined in general terms as the upper aquifer. See Documentation Records p. 5. The disputed issue regards the proper definition of the "target area" of the aquifer of concern, i.e., the area of the aquifer of concern that should be used for purposes of calculating the "Population Served" component of the Targets score.

The target area for the proposed Kysor HRS scoring was defined as all portions of the upper aquifer within a three mile radius of the Kysor site. The Population Served figure was then calculated through an estimation of the number of people who use wells in the aquifer of concern within that three mile radius.

In defining the target area, no consideration was given to the fact that known hydrogeological data establishes that the contamination will either actually or potentially affect only a small portion of the three mile radius area. This approach exaggerates the Population Served score because it includes groundwater wells and related user populations that are beyond the broadest potential scope of the contaminant plume. In other words, this approach inflates the population figure beyond the actual population at risk.

In doing so, this approach violates and undermines the provisions, policies and purposes of CERCLA and the NPL. Section 105(8) of CERCLA is the statutory provision requiring the promulgation of the NPL and HRS. That section specifically states that the NPL and HRS "[c]riteria and priorities shall be based upon relative risk or danger to public health . . . taking into account to the extent possible the population at risk" CERCLA § 105(8)(A) (emphasis supplied).

This language clearly requires that the NPL rank sites according to relative risk, and that the HRS include the population at risk as a major component in the scoring system. The Introduction to the HRS expressly acknowledges those requirements: "In order to set the priorities, CERCLA requires that criteria be established based on relative risk or danger, taking into account the population at risk" The Introduction to the HRS further acknowledges that the HRS is intended to allow EPA to "identify those releases . . . that pose the greatest hazard to humans or the environment." The Population Served score, therefore, is intended to measure the population at risk. HRS Text, "Population Served by Groundwater."

In short, CERCLA employs the NPL as a priority list based on relative risk of human and environmental injury. CERCLA requires that the HRS quantify potential risk using specific criteria including "population at risk." In contrast, nothing in CERCLA authorizes EPA to use non-risk oriented factors in applying the HRS.

This risk based scoring and ranking system was adopted so that limited Superfund monies would be directed to the most dangerous sites. For example, only those sites listed on the NPL are eligible for Superfund financed remedial actions. NCP § 300.68. In addition, the NPL influences EPA enforcement decisions. A misallocation of Superfund monies and enforcement efforts will occur to the extent that HRS scoring overstates the actual risk of a site. In effect, such scoring errors defeat the basic purposes of CERCLA.

That exact result will occur in this instance unless the proposed Kysor HRS scoring is revised to reflect only the actual population at risk. The aquifer target area used in the Kysor scoring included people who are at no conceivable risk (as discussed further below). The EPA contractor's approach thereby violated the CERCLA requirement that HRS scoring quantify relative risk. In simple terms, the EPA contractor's approach results in an erroneous score that would cause a misallocation of limited CERCLA resources if this site is included on the NPL.

A proper scoring for the Kysor site requires a redefinition of the target area of the aquifer of concern. The HRS provides no formal definition of aquifer of concern nor any justification for use of the three mile target area. Reference to CERCLA in constructing the proper definition is required since CERCLA is the controlling statute. In contrast to the EPA contractor's approach, CERCLA indicates that the target area definition must

comply with CERCLA's express requirement that HRS scoring reflect the population at risk. Since population at risk is the factor being measured, it is inescapably logical that the target area should be defined as that portion of the affected aquifer that is realistically exposed to potential contamination.

In this instance, known hydrogeological data establishes that the area of potential contamination is relatively small. The only defensible justification for not considering hydrogeological data is that such information typically is not available when HRS scoring is conducted. In fact, EPA has stated this precise reason as the basis for the use of the three mile radius target area. 49 Federal Register 37070, 37077 (September 21, 1984); 47 Federal Register 31180, 31190 (July 16, 1982). If the information is available, however, there is no statutory basis for refusing to use it. In fact, CERCLA's requirement of a risk based scoring system seems to compel the use of such information since it is particularly relevant in measuring the extent of the risk. Ignoring the information when it is available is directly contrary to CERCLA because it unjustifiably exaggerates the risk and HRS score for a site.

Moreover, ignoring such data in defining the target area is inconsistent with the fact that the HRS considers known hydrogeological data in calculating the nearest groundwater well. (See Figure 6 of the HRS, in which it is explained that a well should not be considered if known hydrogeological data indicate that the well will not be affected.) Similarly, when the NPL and

HRS were promulgated, EPA specifically stated that "geohydrological data on known aquifer interruptions may be used to show that potential targets which are located within three miles do not draw from the affected aquifer." 47 Federal Register 31180, 31190 (July 16, 1982). There is no logical basis for using the information in one instance but not the other. Instead, consistent application of the HRS requires that hydrogeological data, if available, be used for all relevant purposes.

Use of available groundwater data is also consistent with the NCP's cost effectiveness requirement. If available data is ignored, remedial action costs may be incurred unnecessarily to duplicate existing information or to investigate a site that does not merit the expenditure.

As noted above, there is established groundwater data concerning the area. The data is sufficient to define the limited area of the aquifer of concern that is exposed to potential contamination. That data is explained and analyzed in the attached letter (Ex. A) prepared by Fishbeck, Thompson, Carr & Huber ("Fishbeck"). Fishbeck is a professional engineering firm with extensive experience in environmental and geotechnical investigations. Fishbeck has been engaged in hydrogeological work at the Kysor site during the past four years. In addition, the individual author of the letter was involved in extensive hydrogeological studies in the Cadillac, Michigan area prior to the Kysor work. His prior work included involvement in the groundwater studies for the initial development of the Cadillac municipal well field.

The data indicate that the target area of the aquifer of concern can be defined legitimately in several ways depending on how precisely the groundwater data is applied. Each method, as described below, results in a more limited and more logically defined area than under the EPA contractor's approach. Moreover, each area is more consistent with CERCLA's mandate that HRS scoring reflect the population at risk. The scope and underlying justification of each area are described below. The population counts for each and the HRS scoring are then discussed in the next section.

The target area that is consistent with CERCLA, since it provides the most precise and accurate measure of the population at risk, is defined as follows: that area of the aquifer of concern that is bounded on the South by an East-West line through the Kysor site, on the North by the three mile radius line around the Kysor site, and laterally on the East and West by North-South lines that are, respectively, three quarters of one mile East and West of a North-South line through the former waste pit at the Kysor site. The groundwater data (as described more fully in the Fishbeck letter) fully supports this definition. The groundwater flows North from the Kysor site. Consistent with the groundwater flow, the contaminant plume is migrating North of the former waste pit. Areas to the South of the Kysor site, therefore, are not threatened with contamination. (See Fishbeck letter, p. 2 discussion of Area II.) The East-West boundaries are justified on the basis that the contaminant plume is not and will not spread laterally and there is no reasonable possibility that contaminants will be drawn more than 3/4 of a mile to the East or West within

the three mile radius area. (See Fishbeck letter, pp. 3-4.) The North boundary is defined as provided in the HRS. This target area is hereinafter referred to as "Target Area A."

An alternative target area definition involves a less precise application of the groundwater data. This definition employs the same boundaries described above except that the South boundary is formed by the Eastern shore of Lake Mitchell, the Northern shore of Lake Cadillac, and the Ann Arbor Railroad tracks that run East/Southeast from the Northeast corner of Lake Cadillac. Areas adjacent to the lake shores are not included. The bases for the North, East and West boundaries are the same as described above. The South boundary is based on groundwater data establishing that Lake Mitchell and Lake Cadillac act as hydraulic interruptions that prevent groundwater flow from the North to the South. (See Fishbeck letter, p. 2 discussion of Area I.) The shore areas are not included because the lakes provide extensive recharge to adjacent wells. The recharge effect is sufficient to create an effective hydraulic barrier between the wells in the lake shore area and the contaminated groundwater to the North. (See Fishbeck letter, p. 2 discussion of Area II.) This target area definition is less consistent with CERCLA than Area A because it ignores the data establishing that areas South of Kysor are not at risk. This target area is hereinafter referred to as "Target Area B."

The final target area consists, in general terms, of the three mile radius area North of Lake Mitchell and Lake Cadillac, but not including the areas adjacent to the East shore

of Lake Mitchell and the North shore of Lake Cadillac. The precise boundaries are as follows: on the North, East, and West, the boundary is formed by the three mile radius line around the Kysor site, and on the South the boundary is formed by the East shore of Lake Mitchell, the North shore of Lake Cadillac, and the Ann Arbor Railroad tracks that run East/Southeast from the Northeast corner of Lake Cadillac. As noted, the areas adjacent to the lake shores are not included. The East, West and North boundaries are based on the HRS' use of the three mile radius line. The South boundary is the same as for Target Area B and the same justifications apply. This target area definition is the least consistent with CERCLA because it includes a large area that is not realistically exposed to contamination and provides the least accurate measure of the population at risk. This target area is hereinafter referred to as "Target Area C."

B. Population Counts and HRS Scoring.

This section will discuss the population count and resulting HRS scoring for each Target Area as defined above. The Target Areas are discussed in the same order as in the previous section.

Kysor used the same methodology to calculate the population counts for each Target Area. Kysor's methodology was designed to comply with the HRS requirement that the Population Served score reflect the number of people who actually use groundwater. In contrast to the EPA contractor's rough estimations and imprecise

techniques, Kysor made extensive efforts to obtain an accurate count of the actual population using groundwater in each Target Area. Kysor made the count through the use of the following sources: (1) aerial photographs (scale 1:400) of the relevant area; (2) Polk's 1985 Cadillac, Michigan City and Rural Route Directory, which identifies each building listed as a residence or a business; (3) City of Cadillac maps that identify individual addresses and whether each uses city water or well water; and (4) census data for Wexford County and Cadillac, Michigan, provided by Data Research Center, Inc. of Traverse City, Michigan, including 1980 U.S. Census data and 1985 census data compiled by Data Research Center. Kysor applied each source independently and then cross-referenced between each source for verification. A copy of Data Research Center, Inc.'s letter listing the census data is attached as Exhibit B.

Kysor made the counts using the following procedure. For areas inside the city limits, the city map was used to identify all buildings using well water. The aerial photos were used to count buildings in the township areas outside the city. The Polk Guide was then used to identify whether each building was a residence or business. Each occupied building outside the city limits was assumed to use groundwater. For residences, a population figure of 2.76 persons/household was used. This figure is the average number of persons per household for Wexford County, Michigan, as contained in the 1980 U.S. Census. See Ex. B. Use of the Wexford County figure obviously provides a much more accurate result than using the national average as in the HRS.

The results of the population counts and HRS scoring are as follows:

In Target Area A, there are 26 residences in the city using groundwater. There are 182 residences in the township areas using groundwater. The total number of residences is 208. Applying the 2.76 persons/household average for Wexford County, the total number of inhabitants in the residences is 574. There are 21 businesses using groundwater within Target Area A. Those businesses have a total of 134 employees. The total number of residents and employees is 708. Using that population figure, the HRS assigned value for the Population Served is "2". In conjunction with a "distance to nearest well" assigned value of "4", the matrix score for Population Served is "20". Applying that matrix score in the HRS formula with no other changes from the previous scoring, the HRS score is 22.37 (as compared to the proposed HRS scoring of 33.94).

In Target Area B, there are an additional 123 residences using groundwater in the area added to Area A between Kysor and Lake Mitchell and Lake Cadillac. The total number of residences in Target Area B is 331. The total number of inhabitants, using the 2.76 persons average, is 913.6. The total number of business employees using groundwater is 134, the same as for Area A. The total number of people using groundwater in Area B is 1,047.6. Substituting that population figure into the HRS formula, and making no other changes

in EPA's scoring, the HRS score is 30.08. While Kysor's population counts are much more precise than EPA's, they are not exact. Since the 1,047 population count for this area is very close to the 1,000 figure used as a scoring cut-off, the population count for this area merits further investigation if this target area is used.

Target Area C, which extends the East and West boundaries to the three mile radius line, contains 731 residences using groundwater. The total number of residential inhabitants in the Area is 2,017.6. The Area also includes 121 businesses using groundwater. Kysor does not have an actual count of the number of employees in this area, but the average number of employees per business in Areas A and B was 6.4. Many of the businesses in Area C are in the rural township areas and presumably have fewer employees than the businesses in Areas A and B. The same 6.4 employee per business average will be applied, however, for purposes of consistency. Using that figure, the total number of employees is 774.4. The total population using groundwater is 2,792. The HRS score using that population figure is 30.08, the same as for Area B.

Kysor believes that Target Area A should be employed in rescoring the Kysor site. This Area, consistent with CERCLA's express requirements for HRS scoring, provides the best measure

of the population at risk. The resulting HRS score, therefore, is the most accurate estimate of the relative risk of the Kysor site. Since the entire NPL procedure is based on relative risk, the HRS score for Area A should be used. Using that HRS score, the Kysor site is not eligible for NPL listing. That result is logical and proper in light of the relatively small population that is actually exposed to potential contamination.

Target Areas B and C are less precise than Area A. Both Areas, however, provide more accurate measures of the population at risk than the arbitrary three mile radius employed in the HRS. Though not optimal, Areas B and C are more legitimate than the approach actually used by the EPA contractor in the Kysor scoring.

Kysor recognizes that EPA may be concerned that using groundwater data in this instance will raise questions about the accuracy of scores calculated for other sites. The possibility that errors were made on other sites, however, does not justify failing to correct the error that has been made on the Kysor site. All of the policies underlying CERCLA, including the cost effectiveness requirement in the NCP, require EPA to use all available information that will improve the accuracy of the scoring.

II. Erroneous Site Identification.

As explained in detail above, Kysor objects to the EPA contractor's broad definition of the target area of the aquifer

of concern. Kysor strongly believes that the known groundwater data must be used to redefine the target aquifer.

In addition, Kysor believes that the site name should refer to the general area of the contamination rather than the Kysor site specifically. Even if the site name is revised to reflect a more general area, Kysor believes that the known groundwater data should be used to reconsider and reduce the HRS score.

Kysor's objection regarding the site name is premised on the fact that at least three non-Kysor facilities are allegedly contributing to the groundwater contamination in the target area of the aquifer. If the broad target area definition is used, the other facilities are affecting the same aquifer allegedly affected by Kysor. In addition, the other facilities threaten the same target population under the broad definition.

CERCLA § 104(d)(4) authorizes EPA to treat two or more noncontiguous facilities as one if the facilities are reasonably related on the basis of geography or threat to the environment. EPA has stated that it will consolidate noncontiguous facilities for site identification purposes under appropriate circumstances. E.g., 49 Federal Register 37070, 37076 (September 21, 1984) (NPL update); 48 Federal Register 40658, 40663 (September 8, 1983) (initial promulgation of NPL). Relevant factors are whether contamination from the separate sites is threatening the same aquifer and whether the target population is essentially the same or

substantially overlapping. Id. Those factors certainly exist under the broad target area definition.

The Michigan Department of Natural Resources ("DNR") recognized that multiple sources of contamination may exist when the DNR scored the area under the Michigan equivalent of CERCLA. The DNR has listed the area as the "Cadillac Area Groundwater Contamination" on the Michigan Act 307 Priority List.

In addition to the factors identified above, fundamental fairness indicates that a general site name is required. It would be manifestly unfair for EPA to use the broad target area, which includes contamination from several alleged sources, and yet focus solely on Kysor, one alleged responsible party, in the NPL listing process. EPA cannot use a very broad approach in defining the target aquifer but then use an overly narrow approach in defining the site.

The consequences of employing inconsistent approaches are significant. EPA will misidentify remedies and PRPs if the site area is not properly matched to the area of contamination. Similar results will occur in this case unless EPA either narrows its definition of the aquifer or broadens its definition of the site.

The HRS scoring will still require reevaluation if the site name is revised. The known groundwater data can be applied in improving the accuracy of the score regardless of the site name.

III. Conclusion.

Kysor's comments focus on two objections to EPA's proposal to add Kysor's facility to the NPL: erroneous HRS scoring and erroneous site identification.

Kysor believes that the proposed HRS scoring is overstated because of errors in calculating the population at risk. Known hydrogeological data was ignored when the population at risk figure was estimated for the Kysor site. As a result, the population at risk figure was exaggerated and the proposed HRS score erroneously overstates the relative risk of the site. If properly scored, the Kysor site is not eligible for NPL listing.

Failure to consider available groundwater data violates the express provisions and policies of CERCLA and the NPL scheme. CERCLA expressly requires that the HRS quantify relative risk and use population at risk as a major scoring factor. The HRS must measure risk since the NPL is a priority system based entirely on relative risk. Groundwater data is critically relevant in making an informed and accurate measure of population at risk. Ignoring such data when it is available creates an unnecessarily inaccurate score in direct contravention of the CERCLA NPL scheme.

The potential negative consequences of ignoring the data significantly outweigh any purported difficulty in applying the data. A misallocation of limited Superfund monies and enforcement efforts will occur unless HRS scores accurately reflect

relative risk. Moreover, ignoring the data will result in duplicative expenditures to gather existing data or unnecessary expenditures for response costs at sites that do not merit CERCLA response. These consequences vary in significance, but they all share the common trait of violating CERCLA's cost effectiveness requirement.

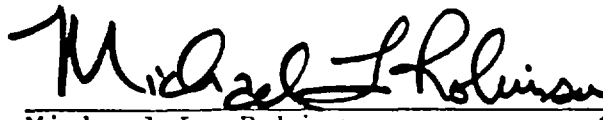
In light of the errors in the proposed scoring, Kysor requests that EPA rescore the Kysor site and employ the known hydrogeological data in doing so. As described in detail above, Kysor believes that a proper application of that data results in an HRS score of 22.37. That score is the most accurate measure of the risk of the Kysor site and it should be used for NPL purposes. Based on that score, the Kysor site should not be listed.

Kysor also objects to the site identification aspect of the proposal. The proposed listing would specifically use Kysor's facility as the site name even though there are several other facilities allegedly causing contamination in the same aquifer as it is broadly defined in the proposed scoring. If a broad approach is to be used in defining the target aquifer, then a similarly broad approach must be used in identifying the sources of contamination and in listing the site name. Otherwise, EPA will misidentify appropriate remedies and PRPs, as well as focus unfairly on particular PRPs who are only partially responsible if at all. Accordingly, Kysor requests that EPA use a general site name designating the area of the contamination. The revised site name should not focus on Kysor.

Finally, the groundwater data should be considered in rescoring the area even if a revised site name is used. Use of the data in accurately measuring the population at risk is required in all instances.

Kysor appreciates EPA's consideration of these comments.

Respectfully submitted,


Michael L. Robinson 

and


John D. Dunn

Exhibit A

fishbeck, thompson, carr & huber / fairbrother

engineers & scientists

Walter W. Meinert, P.E.
Principal, Vice President

November 13, 1985
Project No. F81817/572M

Mr. Michael L. Robinson
Warner, Norcross & Judd
Attorneys at Law
900 Old Kent Bank Building
One Vandenberg Center
Grand Rapids, MI 49503

Re: Kysor of Cadillac, Groundwater

Dear Mr. Robinson:

As per our previous discussion, it is our understanding that the Kysor Corporation, manufacturing facility in Cadillac, Michigan has been placed on the National Priorities List (NPL). The inclusion of the Kysor facility on the NPL was determined by the EPA's scoring system as outlined under the Hazard Ranking System (HRS).

The ranking system score represents an analysis of the probability and magnitude of harm to the human population or sensitive environment from exposure to hazardous substances as a result of contamination of groundwater, surface water or air.

The preamble of the NPL contains discussion that the computation of population served by groundwater from the aquifer of concern should include all population within a three mile radius of the facility. The EPA's approach was based on the difficulty of predicting precisely the movement of groundwater because sufficient data is unavailable when HRS scoring takes place.

Previous hydrogeological investigations conducted for the Kysor Corporation have defined the direction of groundwater movement within the aquifer of concern. These studies have also defined the regional and local geology and hydrogeology.

It is our intent, with this correspondence, to identify interruptions or discontinuities within the aquifer of concern, in the vicinity of the Kysor facility. These discontinuities reduce the extent of population served. This reduction may be divided into three general areas as follows:

Mr. Michael L. Robinson

Page 2

November 13, 1985

AREA I

Area I includes that portion of population situated south of Lakes Cadillac and Mitchell and contained within the three mile radius of the facility.

Lakes Cadillac and Mitchell are both perched bodies of water situated above the natural groundwater surface. The elevation of Lake Cadillac is 1,290 feet msl. The elevation of Lake Mitchell is 1,291 feet msl. The groundwater elevation at a point approximately 800 feet south of Lake Cadillac is 1,275 feet msl. The groundwater elevation near the Kysor facility is also at elevation 1,275 feet msl. The degree of recharge from these lakes to the underlying aquifer is less than perfect (100 percent). The previous hydrogeological studies have shown that the shallow subsurface materials have favorable permeabilities.

If a perfect hydraulic connection exists between the lakes and the groundwater, along with the favorable permeabilities of the shallow materials, the lake elevation would agree closely with that of the groundwater surface. The relatively flat groundwater surface in the area of Lake Cadillac, however, suggests a degree of recharge to the groundwater. This same condition would also exist in the vicinity of Lake Mitchell.

The aquifer recharge from the lakes would, therefore, provide a hydraulic boundary between wells located south of the lakes and that portion of the aquifer near the Kysor facility. The previous hydrogeological investigations mentioned above have shown the direction of groundwater movement to be to the north. The amount of recharge provided by the lakes would not permit wells south of the lakes to induce groundwater movement from the Kysor plant area.

AREA II

Area II includes that population between the Kysor plant and Lake Cadillac. This area to the south is upgradient from the plant. As a result of the groundwater flow gradients, we would not expect the movement of contaminants in this southerly upgradient direction. This area is also zoned residential and therefore, it is highly unlikely that a well of sufficient capacity to induce the movement of contaminants to the south, would ever be drilled in this area. The population in this area south of the Kysor plant therefore should not be included in the HRS.

Area II also includes that population located along the northerly edge of Lake Cadillac and the easterly edge of Lake Mitchell. The cone of depression created about wells in this area would be vastly limited in the downgradient direction, again due to the volume of recharge received from the lakes. This recharge effect will provide an adequate hydraulic boundary condition for all wells adjacent to either Lake Cadillac or Lake Mitchell. The net effect would restrict the movement of groundwater and contaminants from the Kysor plant area toward these wells.

Mr. Michael L. Robinson

Page 3

November 13, 1985

AREA III

The hydrogeological studies for Kysor Corporation completed in March, 1982 and August, 1983 have defined the direction of groundwater movement and the hydraulic gradient. The lateral limits of contamination have also been defined. As part of the development of the city of Cadillac well field in 1959, pumping tests were performed to evaluate the hydraulic characteristics of the aquifer of concern. These characteristics are given by the coefficient of transmissivity "T" and the coefficient of storage "S". Using values as provided in a report entitled "Water Improvement Report" city of Cadillac, Michigan, October 1959, by Jones, Henry & Williams Consulting Engineers, we have attempted to determine the rate of pumping necessary to significantly divert the normal northerly direction of groundwater movement. The hydraulic characteristics used were $T = 230,000$ gallons per day per foot and $S = 0.11$.

The solution of this mathematical evaluation requires certain assumptions. These assumptions deal with the magnitude of hydraulic impact, in the form of drawdown of water levels, necessary to divert water and contaminants from their normal direction of movement. Other assumptions are the distance from the source that a predetermined drawdown occurs and the time required to attain this drawdown.

To err on the side of conservatism, we have elected to learn the pumping rate required to lower the normal groundwater level 1.0 feet below a source of contamination when pumping at a distance of three fourths of a mile from a source of contamination. We have also elected to use a time period of five years of constant pumping to attain this drawdown.

The above hydraulic characteristics, together with the assumptions provided, have been utilized in the "Theis" nonequilibrium equation. C.V. Theis, U.S. Geological Survey developed the formula in 1935. When a well penetrating an extensive aquifer is pumped at a constant rate, the influence of the discharge extends outward with time. The "Theis" equation was the first to take account of the effect of time of pumping on well yield. In its simplest form, the Theis equation is:

$$s = \frac{114.6 Q}{T} W(u)$$

where:

s = drawdown in feet at any point in the vicinity of a well discharging at a constant rate.

Q = pumping rate in gallons per minute.

T = coefficient of transmissivity of the aquifer, in gallons per day per foot.

W(u) is read "well function of u" and is short for the exponential integral describing u.

Mr. Michael L. Robinson

Page 4

November 13, 1985

We have used the Theis formula to determine what pumping rate would be necessary to create a drawdown at a given distance and time as discussed above. The solution to this equation results in a pumping rate of 469 gallons per minute. This pumping rate would, therefore, be necessary at a point three fourths of a mile from a source of contamination to create a drawdown of the groundwater surface beneath the source of contamination of 1.0 feet. Furthermore, this pumping rate must be constant (with no interruptions) for a period of five years. This drawdown beneath a source of contamination could theoretically create a diversion of the direction of normal groundwater movement in both an east and west direction from the Kysor plant area. The area located east of the plant is within an urban neighborhood which is supplied by the municipal water system. The possibility of a high capacity well being drilled in this area is highly remote. We, therefore, feel that all population beyond three fourths of a mile east of the Kysor plant should not be considered in the site scoring system. The area this same distance mile west of the Kysor plant is shown on the U.S.G.S. topo map as low swampy terrain. As discussed above, it is highly remote that a high capacity well would be developed in this area. Without any means to divert the normal direction of groundwater movement towards the west, we feel that all population beyond three fourths of a mile west of the Kysor plant also be excluded from the site scoring procedure.

We believe that sufficient information is available on both the local and regional hydrogeology of the subject area to permit a calculation of the population at risk in accord with the above analysis.

If you have any questions regarding this interpretation, we would be pleased to discuss them with you.

Sincerely,

FISHBECK, THOMPSON, CARR & HUBER, INC.



Walter W. Meinert, P.E.

rb

Exhibit B

DataResearch

November 14, 1985

Warner, Norcross & Judd
Mr. John Dunn
900 Old Kent Building
Grand Rapids, MI 49503

Dear Mr. Dunn:

As per our phone conversation of November 8, 1985 I have listed below the information you requested.

Population

<u>Township/City</u>	<u>1980 (1)</u>	<u>1985 (2)</u>	<u>1980-1985 Percent Change</u>
Cadillac City	10,199	9,674	-5.1%
Cherry Grove	1,517	1,439	-5.1%
Clam Lake	1,658	1,588	-4.2%
Haring	2,523	2,453	-2.8%
Selma	1,289	1,249	-3.1%

Household

	<u>1970 (1)</u>	<u>1980 (1)</u>	<u>Percent Change</u>
Persons per House- hold (Wexford County)	3.16	2.76	-12.7%

(1) United States Bureau of Census

(2) Data Research Center, Inc.

Thank you for your call.

If we can be of further service, please call Data Research.

Sincerely,


Paul E. Opplinger
Executive Director

PEO:dwg

Enclosures

IDENTIFICATION OF TERMS

1. ARARS - Applicable or Relevant and Appropriate Requirements. U.S. EPA's term for the federal and state requirements that CERCLA remedial actions must comply with pursuant to CERCLA § 121(d).
2. CERCLA - The Comprehensive Environmental Response, Compensation and Liability Act, 42 U.S.C. § 9601, et seq.
3. MCL - Maximum Contaminant Level. The enforceable drinking water standards promulgated by U.S. EPA pursuant to the Safe Drinking Water Act, 42 U.S.C. § 300f, et seq. These standards apply to public drinking water systems.
4. MCLG - Maximum Contaminant Level Goal. The unenforceable health goals developed by U.S. EPA pursuant to the Safe Drinking Water Act, 42 U.S.C. § 300f, et seq. These goals are used in setting the MCLs.
5. NCP - National Contingency Plan, 40 C.F.R. § 300.1, et seq. The regulations promulgated by U.S. EPA pursuant to CERCLA to govern CERCLA response actions.
6. SDWA - Safe Drinking Water Act, 42 U.S.C. § 300f, et seq.
7. SMCL - Secondary Maximum Contaminant Level. The unenforceable drinking water criteria developed by U.S. EPA pursuant to the Safe Drinking Water Act, 42 U.S.C. § 300f, et seq. These criteria relate to aesthetic qualities such as odor and appearance.

Exhibit C

Walter W. Meinert, P.E.
Principal, Vice President

August 25, 1989
Project No. F88170

fishbeck, thompson, carr & huber
engineers & scientists

Mr. John D. Dunn
Warner, Norcross & Judd
900 Old Kent Bank Building
One Vandenberg Center
Grand Rapids, MI 49503

Re: Kysor Industrial Corporation RI/FS

Dear John:

As a supplement to comments on the RI/FS and proposed plan for the Cadillac Industrial Park, we offer the following discussion on the local geology.

Considerable review of the site stratigraphy has been made by E.C. Jordan during the RI. The RI report states that the contamination plumes located west of the well field are situated downgradient and do not pose a significant threat, if any, to the municipal well field.

We have reviewed the drilling logs of the wells within the municipal well field in an attempt to document the existence and thickness of protective clay layers overlying the aquifer which supplies the City of Cadillac well field. We have tabulated below the depths and thickness of clay layers at the location of six municipal production wells.

<u>Well No.</u>	<u>Clay Interval (feet)</u>	<u>Total Thickness (feet)</u>
1	175-182 224-232 244-251	22
2	166-176 226-239	23
3	158-177 222-229 237-244	33
4	150-162 170-171 236-243	20
5	144-150 162-165 222-224 229-239	21
6	140-148 171-172 226-248	31

Mr. John D. Dunn
Page 2
August 25, 1989

On August 3, 1983, Fishbeck, Thompson, Carr & Huber, Inc. (FTC&H) performed a pumping test at the municipal well field to determine the hydraulic connection between the lower aquifer, which supplies the municipal well field, and the intermediate aquifer, which contains detectable concentrations of volatile organic compounds. This test involved pumping a production well (No. 7) continuously for 24 hours at a rate of 1,400 gallons per minute. Water level measurements were taken in well No. 7 and a shallow observation well located 50 feet from the pumping well. Whereas approximately 16 feet of drawdown occurred within the pumping well, there was no discernible drawdown within the shallow observation well. A plot of the water level measurements taken during this test are included herewith.

A review of E.C. Jordan well records together with logs of the municipal wells suggests that the protective clay layer is thickening in a westerly direction in the immediate area of the well field. This further supports the conclusions contained in the RI/FS report.

Sincerely,

FISHBECK, THOMPSON, CARR & HUBER, INC.

A handwritten signature in cursive script, reading "Walter W. Meinert".

Walter W. Meinert, P.E.

rb

Enclosure

TIME (MINUTES)

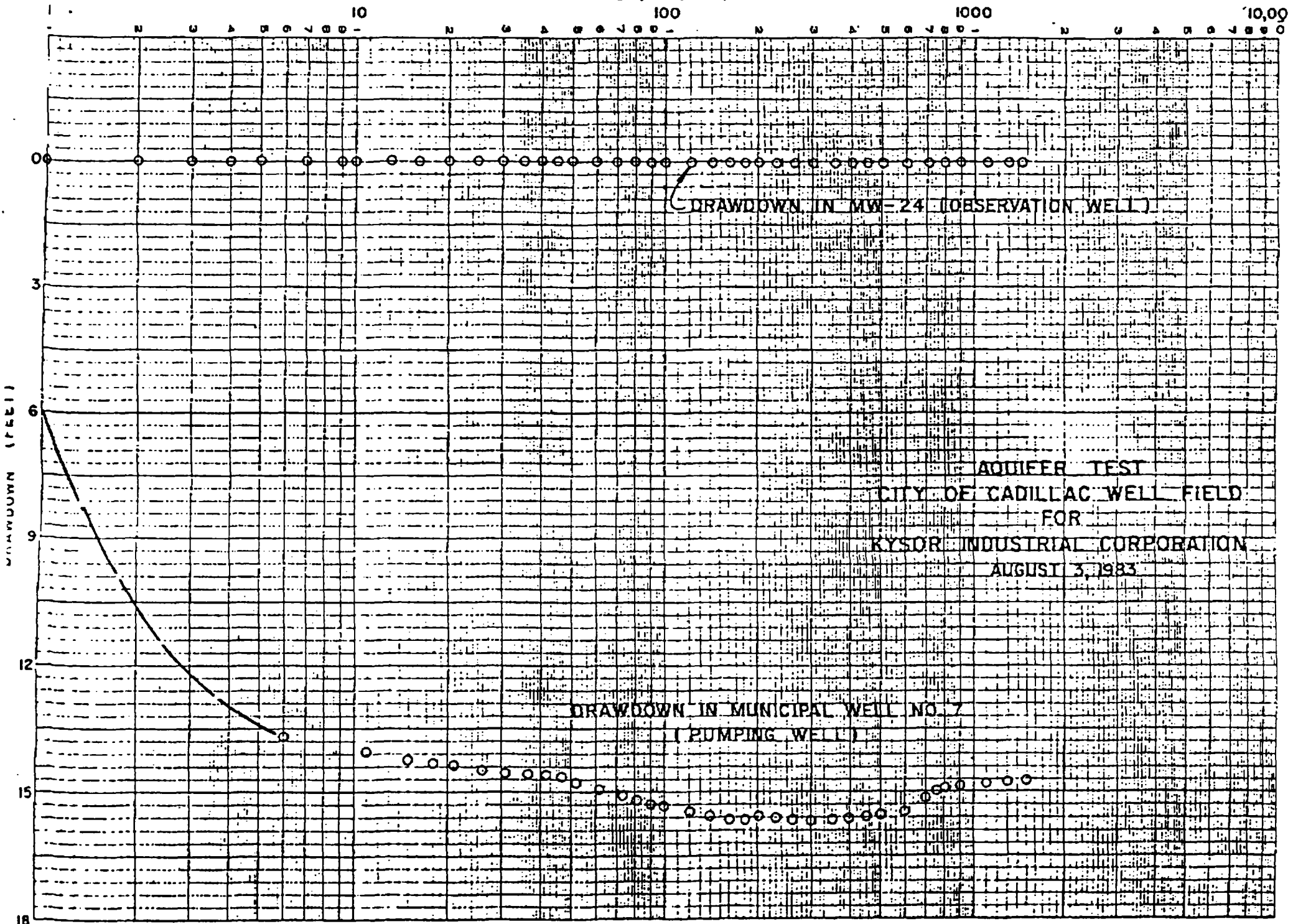


Exhibit D

(1)

SUMMARY of REMEDIAL ALTERNATIVE SELECTION

**U.S. AVIEX SITE
NILES, MICHIGAN**

August 9, 1988

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SUMMARY of REMEDIAL ALTERNATIVE SELECTION

U.S. AVIEX SITE Niles, Michigan

I. Site Location and Description

The U.S. Aviex Site is located at 1056 Huntly Road, City of Niles, Howard Township, Cass County, Michigan in a residential neighborhood made up of essentially single family homes. The nearest homes are within 100 feet south, east and north of the approximately six acre property. Minor agricultural and horticultural activities take place in the general vicinity of the site. There are three schools approximately 2.0 miles west of the U.S. Aviex site. Two schools are located about 1.5 miles northwest of the site. There are no storm or sanitary sewers nearby. Surface runoff from the U.S. Aviex site and nearby areas flows toward the Bame-Huntly drain and subsequently into Brandywine Creek, which begins about 0.5 mile east of the site. Brandywine Creek flows into the St. Joseph River in Niles. The St. Joseph River flows into Lake Michigan at St. Joseph, Michigan. See Figure 1 for a map of the U.S. Aviex plant site.

Baron Lake, a seepage lake with no surface inlets or outlets, is about 0.5 miles northeast of the site. The lake has a surface area of 199 acres and a maximum depth of 28 feet. There is a small wetlands west of the lake.

Site topography is generally level, with occasional low rises and gentle depressions. Very generally, the surface elevation rises 15-18 feet per mile from east to west in the vicinity of the site.

The site is located in an area of glacial deposits. Two broad types of deposits occur: glacial outwash and ice contact outwash. Typically these deposits are stratified and exhibit varying degrees of sorting between individual units, with considerable lateral variation. The upper unit of sand and gravel contains discontinuous clay layers and lenses; within this unit the vadose zone is 15 to 45 feet thick.

The water table aquifer is defined by a discontinuous sandy clay layer. The saturated thickness of the aquifer is 70 to 90 feet near the site, increasing to 100 feet in the western portion of the study area. Groundwater flow is southwest to west from the site becoming westerly approximately 0.5 miles downgradient. Transmissivity in the area is between 225,000 gal/day/ft and 31,000 gal/day/ft with a flow velocity of approximately 0.5 ft/day. Limited data indicates the presence of an artesian aquifer below the sandy clay layer with a flow pattern similar to the upper aquifer. The replacement residential wells are placed in this aquifer.

While the plant is no longer in operation, some buildings on the site are used for storage of non-production related materials. The original structures are shown in Figure 2. The structures presently in place

include the west warehouse, the concrete slabs from the middle and east warehouses, the offices, batch room, aerosol filling room, packaging room, truck docks, and the concrete, bituminous, lawn, and most of the gravel surfaces. The tank farm and part of the gravel parking area now consist of uneven ground and a bermed impoundment used for temporary storage of non-hazardous groundwater produced during an aquifer test performed on the site.

II. Site Background

A. Site History and Enforcement Activity

U.S. Aviex produced non-lubricating automotive fluids at the Huntly Road Plant from the early 1960s until late 1978. Activities at the plant included the repackaging of bulk products and the formulation of new products from bulk ingredients.

Sometime during the operation of the plant in the 1960s or 1970s, chlorinated hydrocarbons, including Trichloroethane (TCA), were apparently released into the vadose zone beneath the bituminous pavement south of the process rooms. These soils still contain significant quantities of TCA, Trichloroethylene (TCE), Perchloroethylene (PCE) and dichloro-compounds.

In July 1972, an underground pipeline containing diethyl ether was broken during excavation on the southeast side of the plant site near the west wall of the packaging room. Within a few months, diethyl ether (DEE) was detected in nearby residential wells on Lilac Street. Over the next several years, diethyl ether contamination was identified in downgradient residential wells southwest of the plant as far as Blanchard Street and Almaugus Drive, approximately one half mile away. Under an agreement with the Michigan Department of Public Health (MDPH), U.S. Aviex installed six monitoring wells on the site beginning in 1972 and also began supplying bottled water to homes with contaminated wells. In addition, U.S. Aviex installed new wells in a lower, uncontaminated aquifer for selected residences.

On November 28, 1978 a fire occurred on the site. The large volumes of water used to extinguish the fire washed unknown amounts of chemicals, released from barrels and tanks in the process buildings, onto unpaved areas. Following the fire, chlorinated hydrocarbons were detected in domestic wells as far west as Bame Street.

Early in 1982, the State of Michigan initiated legal action against U.S. Aviex and reached an agreement for a groundwater investigation of the site. On the basis of data developed during this investigation, an on-site groundwater purge and treatment system was designed and installed to contain the contamination outlined in the groundwater investigation. The system pumps up to a combined 200 gal/min through two purge wells from the contaminated upper portion of the aquifer. The water is then treated by air stripping and discharged to the surface water in compliance with a NPDES permit issued by the State of Michigan.

The U.S. Environmental Protection Agency (USEPA) initially became involved in the U.S. Aviox Site in 1984 as a result of its being placed on the Superfund National Priorities List. In 1985, U.S. Aviox agreed to fund a Remedial Investigation/ Feasibility Study (RI/FS) under an Administrative Order on Consent. The goal of the RI/FS was to determine the nature and extent of the threat to the public health or welfare or the environment caused by contamination from the U.S. Aviox plant site and to evaluate alternatives to establish the appropriate remedial actions to prevent or mitigate the migration or release of contamination from the site.

In 1986, U.S. Aviox filed Chapter 11 bankruptcy with the Bankruptcy Court for the Eastern District of Michigan. Funds had been previously set aside by U.S. Aviox to pay for the RI/FS so that the bankruptcy action did not stop the ongoing RI/FS at the site. Over the past two years, the federal government has been actively negotiating with U.S. Aviox and its creditors to reach an agreement to be approved and filed with the Bankruptcy Court, allowing for reimbursement of USEPA's costs of overseeing the RI/FS and some allowance for the anticipated clean-up costs.

Pursuant to Section 121 of SARA, a responsible party is ordinarily afforded the opportunity to negotiate an agreement with USEPA regarding the design and implementation of the selected remedial action. Due to the bankruptcy status of U.S. Aviox, however, the USEPA determined that negotiations for the funding would prove futile since the company does not possess available funds to pay for the clean-up.

Also, in 1985, under Michigan Act 307, the State of Michigan took steps to provide a public water supply and distribution system to the affected area. The system distributes water from the City of Niles public water supply to an estimated 220 homes in the area. The system is now complete and area residences have been connected to it.

B. Current Site Status

Chemicals originating in the soil at the U.S. Aviox plant site have contaminated groundwater beneath the site, formed a plume with the groundwater in the upper aquifer and are flowing off-site in a southwesterly direction. This plume is bounded on the north by a line from the site northwest to near the intersection of Janellen and Carberry Road, on the southeast by an irregular line extending from the site in a southwestern direction, and on the west by Carberry Road. The compounds within the plume have been identified as chlorinated hydrocarbons, diethyl ether and other volatile organic compounds in concentrations in the range of 0.001 mg/L to 5.6 mg/L.

Chemicals in the plume, if unabated, will continue to flow southwest down the hydraulic gradient into farther reaches of the neighborhood. The estimated speed of groundwater movement down the centerline of the plume in the study area is approximately 0.5 feet per day. Table 1 shows concentrations of the indicator compounds at three down-gradient exposure

points as determined by modeling.

Significant quantities of chlorinated hydrocarbons remain in the vadose zone beneath the entrance driveway to a depth of twenty feet near the southeast process buildings. There are about 10,000 cubic yards of contaminated soils in this area and about 1,500 cubic yards of contaminated soil in the truck dock near the west warehouse. Concentrations of volatile organics in these soils range from 0.01 mg/kg to 13.0 mg/kg.

In addition, a source of DEE may remain on the site in the north central area east of the former tank farm (see Figure 2). Additional investigation during the Remedial Design (RD) stage will be necessary to quantify this source and further define the on-site and off-site contamination of the soil and groundwater.

Operation of the purge wells and stripping tower on the site continues as provided for in the 1982 agreement between the State of Michigan and U.S. Aviex. The locations and spheres of influence of the purge wells during typical operation in September 1986 is shown in Figure 3. Containment is essentially complete for the most highly contaminated zone beneath the southeast entrance driveway area as this zone of contamination is within the sphere of influence of the purge wells. However, containment of the contaminants at the west boundary of the property is not complete because the extent of contamination west of the property boundary extends beyond the sphere of influence.

The RI identified twenty-five (25) volatile and semi-volatile organic chemical compounds in the on-site subsurface soils or the on-site and/or off-site groundwater. Table 2 summarizes these contaminants in each medium.

Of these twenty-five (25) volatile and semi-volatile organic chemical compounds, ten (10) were selected as indicator compounds to meet the objectives of the RI. Each indicator compound was selected because it meets one or more of the following criteria:

- * It is a known or suspected carcinogen.
- * It is a compound for which the USEPA has established drinking water Maximum Concentration Levels Goals (MCLGs) or proposed Maximum Concentration Levels (MCLs) or proposed MCLGs.
- * It has a Safe Drinking Water Act Water Quality Criteria value (for drinking water only, no risk consumption).
- * It is a compound identified at more than one sample location.
- * It is a compound which occurs in extremely high concentrations.

The indicator chemical compounds selected for detailed study are:

TABLE I MODELED GROUND-WATER CONCENTRATIONS AT THREE EXPOSURE POINTS

Indicator Chemical	Almaigua St. (0.33 miles)			Carberry Rd. (1 mile)			Municipal Wells (7 miles)		
	Best Estimate, mg/L (a)	Upper Bound, mg/L (b)	Duration of Pulse, years (c)	Best Estimate, mg/L	Upper Bound, mg/L	Duration of Pulse, years	Best Estimate, mg/L	Upper Bound, mg/L	Duration of Pulse, years
Model 1: Instantaneous Release (d)									
Benzene (e)	0.00052	0.001	2-	4.70×10^{-5}	0.00013	15-	2.60×10^{-8}	3.50×10^{-7}	55-
1,1-Dichloroethane	0.09	0.236	7-	0.00094	0.011	44-	-Not Arrived-	-Not Arrived-	-
1,2-Dichloroethane	0.113	0.674	4-57	0.036	0.12	75-	3.50×10^{-6}	0.00011	65-
1,1-Dichloroethylene	0.012	0.038	10-	4.7×10^{-5}	0.00068	55-	-Not Arrived-	-Not Arrived-	-
Ethyl ether	1.07	4.210	4-	0.168	0.733	28-	-Not Arrived-	-Not Arrived-	-
Tetrachloroethylene (e)	0.004	0.009	2-	0.00037	0.0011	15-	2.10×10^{-5}	2.4×10^{-5}	55-
1,1,1-Trichloroethane	0.309	0.680*	2-	0.028	0.082	15-	1.60×10^{-5}	0.00071	55-
Trichloroethylene (e)	0.013	0.029	2-	0.0012	0.004	15-	6.80×10^{-7}	9.0×10^{-5}	55-
Trichlorofluoromethane	0.009	0.195	2-	0.008	0.024	15-	4.50×10^{-6}	6.0×10^{-5}	55-
Model 2: Continuous Release									
Benzene (e)	0.004	0.007	2-	0.00042	0.002	17-	9.38×10^{-8}	1.44×10^{-6}	53-
1,1-Dichloroethane	1.46	2.95	8-	0.00553	0.078	43-	-Not Arrived-	-Not Arrived-	-
1,2-Dichloroethane	2.58	3.5	5-	0.803	3.31	25-	1.48×10^{-5}	0.00051	63-
1,1-Dichloroethylene	0.747	0.497	11-	0.00023	0.0043	53-	-Not Arrived-	-Not Arrived-	-
Ethyl ether	6.41	10	5-	0.563	4.33	29-	-Not Arrived-	-Not Arrived-	-
Tetrachloroethylene (e)	0.039	0.064	2-	0.00369	0.018	17-	8.70×10^{-5}	1.26×10^{-5}	53-
1,1,1-Trichloroethane	2.76	4.55	2-	0.26	1.25	17-	5.86×10^{-6}	0.0009	53-
Trichloroethylene (e)	0.121	0.2	2-	0.011	0.055	17-	2.58×10^{-5}	3.97×10^{-5}	53-
Trichlorofluoromethane	0.773	1.27	2-	0.073	0.35	17-	1.64×10^{-5}	0.00025	53-

(a) Seventy-year mean.

(b) Maximum concentration in 70 years.

(c) The date of the start and end of the arrival concentrations $>10^{-5}$ mg/L at an exposure point. If no second number is reported, pulse continues past 70 years.

(d) Compounds released during spills, but not after purge wells operate.

(e) Calculated using same plume transport properties as 1,1,1-trichloroethane.

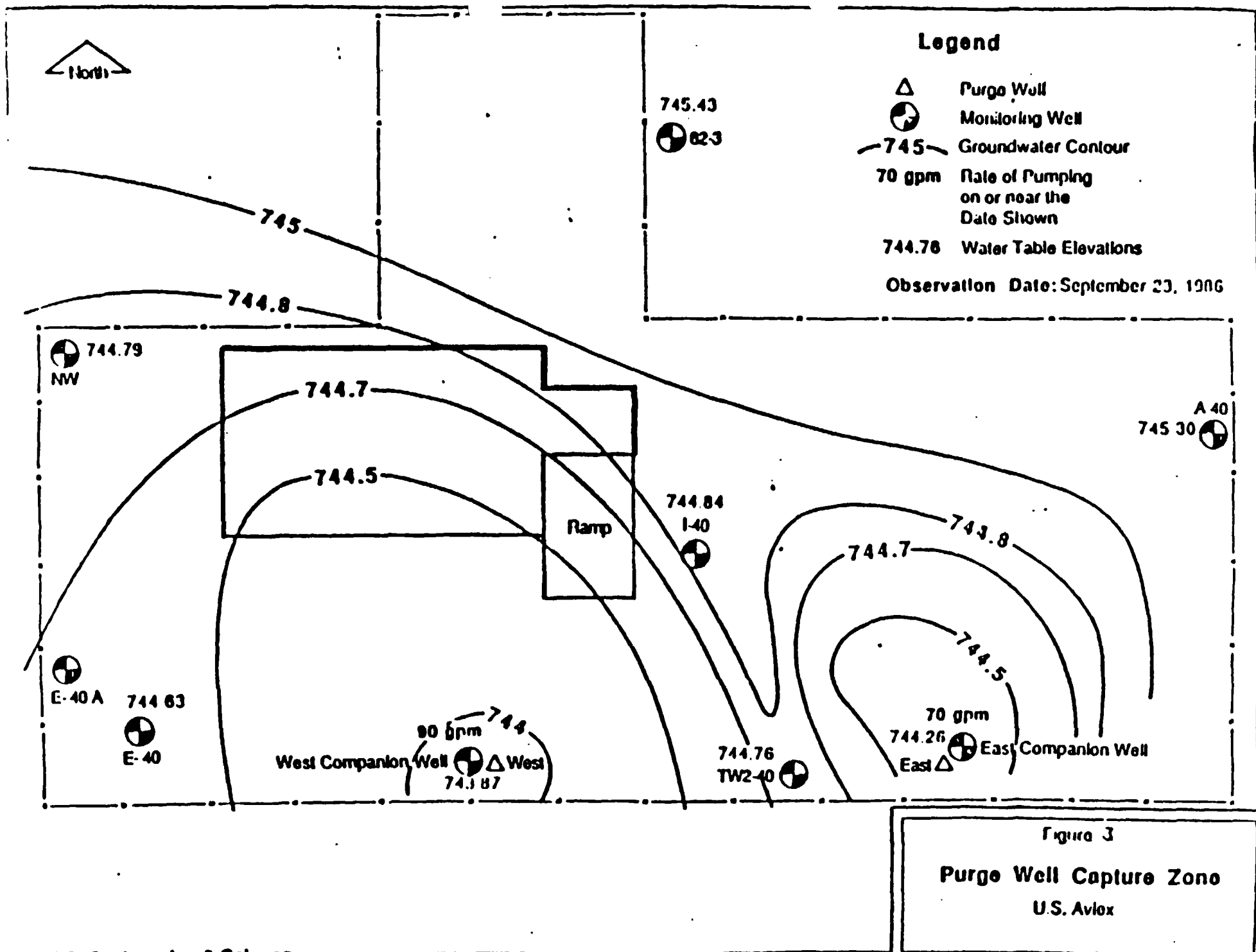


TABLE 2. SUMMARY OF ANALYTICAL RESULTS FOR CHEMICAL COMPOUNDS

Chemical Compounds	On-Site Wells			Off-Site Wells			Soil		
	Range (mg/L)	Mean (mg/L)	(1) N ₁ /N ₂	Range (mg/L)	Mean (mg/L)	N ₁ /N ₂	Range (mg/kg) (2)	Mean (mg/kg)	N ₁ /N ₂
Acetone	0-0.190	0.011	2/20	0-0.33	1.9x10 ⁻⁴	1/171	-	-	-
Benzene	0-0.029	0.003	9/32	0-0.005	3.9x10 ⁻⁵	3/178	-	-	-
Carbon tetrachloride	ND	-	0/22	ND	-	0/178	0-1.230	0.041	1/30
Chloroethane	0-0.050	0.004	3/22	0-0.013	7.3x10 ⁻⁵	1/178	-	-	-
Chloroform	0-0.001	4.5x10 ⁻⁵	1/22	ND	-	0/178	-	-	-
Methyl cyclohexane	0-0.011	0.003	1/4	-	-	0/0	-	-	-
Dichlorodifluoromethane	0-1.000	0.143	2/7	0-0.003	5.0x10 ⁻⁴	1/5	-	-	-
1,1-Dichloroethane	0-3.100	0.366	18/32	0-1.700	0.025	12/178	0-0.120	0.012	5/29
1,2-Dichloroethane	0-1.600	0.089	7/32	0-1.700	0.024	9/178	-	-	-
1,1-Dichloroethylene	0-5.200	0.278	10/32	0-0.330	0.004	4/178	0-0.045	0.005	5/29
Trans-1,2-dichloroethylene	0-0.570	0.059	11/32	0-0.110	1.4x10 ⁻³	4/178	-	-	-
Dichlorofluoromethane	-	-	0	0-0.070	0.019	2/4	-	-	-
Dimethyl disulfide	0-0.003	0.001	1/4	-	-	0/0	-	-	-
Ethyl benzene	0-1.900	0.146	7.29	0-0.090	0.001	6/171	0-0.170	0.008	5/26
Ethyl ether	0-5.700	0.661	15/28	0-4.800	0.036	8/158	-	-	-
Isopropanol	ND	-	0/20	ND	-	0/178	-	-	-
Methylene chloride	0-0.002	9.1x10 ⁻⁵	1/22	0-0.004	3.4x10 ⁻⁵	2/178	-	-	-
Tetrachloroethylene	0-31.00	1.042	11/32	0-0.040	4.1x10 ⁻⁴	4/178	0-10.200	0.761	10/32
Toluene	0-1.600	0.073	1/22	0-0.004	2.8x10 ⁻⁵	2/178	0-0.005	1.7x10 ⁻⁴	1/29
1,1,1-Trichloroethane	0-200.00	15.026	13/32	0-3.00	0.026	18/178	0-23.800	1.631	13/31
Trichloroethylene	0-0.880	0.135	12/29	0-0.130	0-0.001	4/178	0-1.650	0.095	6/31
Trichlorofluoromethane	0-27.000	1.756	12/32	0-0.860	0-0.009	4/168	0-0.160	0.014	1/29
2,2,4-Trimethyl pentane	0-0.005	0.001	1/4	0-0.008	0-0.002	1/4	-	-	-
2,2,4-Trimethyl pentane	-	-	0	0-0.002	0-0.001	1/4	-	-	-
Xylene	0-1.600	0.080	1/20	0-0.010	1.1x10 ⁻⁴	2/178	0-0.010	3.6x10 ⁻⁴	1/28

(1) N₁ = number of samples in which contaminant was detected; N₂ = number of samples analyzed.

(2) -, Compound not analyzed for.

(3) ND = Not detected.

Three of the exposure routes can be quantified. These routes are (1) possible ingestion of contaminated groundwater, (2) direct contact with contaminated on-site soils, and (3) inhalation of volatile emissions from site soils and stack gases.

1. Ingestion Route for potential ground-water and surface-water exposure via drinking water: The most appropriate comparison values are the MCLs of the Safe Drinking Water Act. Since these Federal drinking water standards and criteria are based on lifetime exposures, it is appropriate to use a ratio of long-term water concentration to the standard. Ratios were calculated in two ground-water transport models (instantaneous and continuous release) at 0.33, 1.0 and 2.0 miles from the site. On-site ratios are comparable to those calculated at 0.33 miles. Upper-bound estimates are calculated based on the maximum concentration levels experienced by the three locations in seventy (70) years.

Incremental cancer risks exceeding 10^{-6} indicate potential endangerment (USEPA 1986). Carcinogenic risks from ground-water ingestion exceed 10^{-6} at Almaugus Street (0.33 miles) and Carberry St. (1.0 miles), assuming either continuous or instantaneous release. In the worst case (continuous release) scenario, 1,2-DCA, 1,1-DCE, PCE, TCE and benzene all exceed the critical risk 0.33 miles from the site, and all but benzene and 1,1-DCE exceed this level 1.0 miles from the site. At the municipal wells 2.0 miles from the site, no contaminants of concern are at unacceptable levels now or at the end of the seventy-year period.

Hazard indices for individual indicator chemicals are summed and calculated to provide the total hazard index for the ground-water ingestion route. The noncarcinogenic chronic hazard index exceeds 1.0 only for the worst case scenario 0.33 miles from the site. Two indicator chemical compounds, 1,1-DCE and 1,1-DCA, contribute most heavily to the noncarcinogenic risk index. The chronic hazard index does not exceed 1.0 in seventy years at the more distant locations; the subchronic index does not exceed 1.0 at any location. These conclusions would be the same for resident children even though they have a daily intake about four times that of adults.

2. Soil Contact Route: There are no unacceptable risks for the soil contact route. Since there are no ARARs with which to compare soil contact risk, a comparison was made between the chronic acceptable oral intake values and calculated chemical intakes at the site to determine the potential endangerment to adults by direct contact with on-site soils.

Multiplying chronic daily intakes in mg/kg/day by the oral or inhalation cancer potency factor for the appropriate contaminant, yields the incremental lifetime cancer risk associated with ingestion or inhalation of that contaminant (USEPA 1986). Individual contaminant risks are summed so that the total risk due to carcinogen ingestion or inhalation for a particular route are obtained.

3. Inhalation Route: The potential endangerment from the inhalation

route is low. Since there are no ARARs with which to compare ambient air concentrations resulting from soil volatilization and stack gas emissions daily intakes are calculated using the USEPA's risk assessment guidelines for chemical mixtures (USEPA 1986). Neither subchronic or chronic total noncarcinogenic hazard indices approach 1.0 for either soil volatiles or stack gas. Soil volatiles contribute more contaminants to the air than stack gases but the numbers are very small. Adding the two totals to reflect exposure to both sources has little effect on the index as the totals never exceed 0.05 for best estimates or 0.6 for upper-bound estimates.

Cancer risks from purge well stack emissions are low (2.63×10^{-8}). The highest estimate of incremental cancer risk from inhalation of soil volatiles is 4.76×10^{-7} . Total upper-bound risk, calculated using the highest measured soil concentrations, is 4.72×10^{-6} . 1,1-DCE is the major contributor to this risk. The air inhalation route, therefore, presents a low risk of potential endangerment. The highest cancer risk, estimated from mean soil concentrations of cancer risk from dermal contact with soil, is 1.9×10^{-8} .

III. Community Relations History

Public participation requirements set out by Sections 113 and 117 of CERCLA, as amended, were satisfied during the remedial action process and include the following: Potentially affected persons and the public were notified of the remedial action through two notices published in the local Niles community newspaper; interested persons on the Agency's mailing list received direct mailings; information repositories for public review were established at the Howard Township Hall and the Niles Community Library; the U.S. Aviox Administrative Record was made available to the public and is housed at the Niles Community Library, and Fact Sheets were distributed at public meetings. Two public meetings were held on June 15 and July 12, 1988 to discuss the Proposed Plan and the RI/FS, and accept comments.

A twenty-one (21) day public comment period for the Proposed Plan commenced on June 6 and ended on June 26, 1988. As a result of public interest in the U.S. Aviox remedial action, the public comment period was extended through July 18, 1988. A Responsiveness Summary detailing the Agency's response to significant comments submitted during the comment period, is attached to the ROD.

IV. Scope of Response Action

At the U.S. Aviox site there is a potential public health hazard from exposure to contaminated groundwater. The public health risk results from the possibility of persons in the effected area consuming contaminated groundwater and the long-term potential impact on the City of Niles municipal water supply wells. The purpose of the overall remedial objectives are to provide:

- * On-site control of the source of contamination by minimizing leachate production of contaminated groundwater under the site in

to off-site areas.

- * Off-site management of plume migration. Purging and treating groundwater within the 10^{-6} and/or noncarcinogenic threshold value plume to meet Federal or State drinking water quality standards and criteria will reduce the potential for human exposure to contaminants and reduce the impact on groundwater resources by minimizing migration of off-site contaminants.

To evaluate the remedial action alternatives for the off-site control of groundwater, analytical modeling simulation of the aquifer along with the addition of hypothetical purge wells to capture the contaminant plumes, was done.

Permeability data, developed during the modeling of the aquifer in the RI, was reviewed and a representative value used, along with the typical aquifer thickness, to obtain a transmissivity (T) value of 90,000 gal/day. A storage factor(s) of 0.13, used in RI modeling, and an alternate value of 0.2 were employed along with the T value in an analytical well-field model, developed by EDI, which is based on Theis and Jacobs modified nonequilibrium equation. The relative position of the computer generated capture plumes was nearly the same under the alternative storage factor assumptions.

Other assumptions included in the model are a pumping time period of two weeks (T+14 days) and recharge due to precipitation at twelve inches per year. Several trial runs resulted in alternative scenarios for groundwater capture on-site and off-site using up to five wells, producing in the range of 60 to 180 gal/min.

The remainder of this section discusses the factors considered in the selection of purge well locations, the procedure used to estimate the initial treatment concentrations and the remediation goals for the groundwater and effluent.

A. Selection of Purge Well Locations

The purge wells were located in the areas best suited for evaluating the groundwater capture alternatives and to recover DEE from the semi-isolated shallow groundwater zone area of Purge well 82-3 (See Figure 4.)

Purge well 3A (See Figure 4) will be installed to operate as a supplemental well to the pair of purge wells currently on site. This well will effect continued cleanup of groundwater beneath the site, intercepting chemicals leaching from the vadose zone in the existing entrance driveway area, capturing contaminants at and beyond the southwest corner of the site. The total groundwater flow rate will be approximately 250 gal/min. The groundwater capture zone for this alternative is assumed to be just beyond the site boundaries.

Four purge wells are located in the contaminant plume of toxicological and risk-assigned chemicals up to and including

concentrations at the 10^{-6} level and corresponding toxicological threshold values. The position of the estimated area of groundwater capture and approximate well locations are shown in Figure 5. Purge well 1B, the existing east purge well, is designed to operate at 60 gal/min. and contain the source of contaminants in the entrance driveway area. Purge well 2B, to be located near the intersection of Bame Avenue and Lilac Street, is in the higher concentration zone of the dichlorinated volatile organic compounds (VOCs). The well is designed to operate at 120 gal/min. Purge well 3B is located a little beyond the downgradient limit of the 10^{-6} risk area on Blanchard Street and is intended to produce at 180 gal/min. Purge well 4B is located on the north/south leg of Blanchard Street in the higher concentration zone of contamination of the VOCs and is intended to operate at 120 gal/min.

The total pumpage from the four purge wells is 480 gal/min. When combined with the shallow ether-capture wells on site, the estimated total flow to treatment will be 500 gal/min.

Five purge wells are sited to capture primary groundwater contamination. The position of the estimated area of groundwater capture and approximate well locations are shown in Figure 6. Wells 1C and 2C correspond to Wells 1B and 2B, respectively, and will fulfill the same functions as in the 10^{-6} risk-related alternative. Well 3C is sited approximately 600 feet northwest of its 3B counterpart and off the center of the plume. Well 4C is sited approximately 600 feet northwest of the Yankee Street homes (2379 and 2383) that showed the 1,1,1-TCA. Well 5C is located approximately 350 feet north of Almagus Drive and Lilac Avenue to intercept the low levels of PCE.

Wells 3C and 4C have been shifted to the north of the plume center to effect a continuous groundwater capture zone with Well 5C and maintain an area of influence that extends a sufficient southward distance to capture all the primary plume.

B. Estimation of Initial Treatment Concentration

Treatment influent and effluent data from the existing operational purge and treat system were used to estimate initial concentrations under the groundwater capture alternatives. Concentrations insignificant for developing treatment alternatives were not considered and included very low levels of chlorinated organics and DEE.

The estimation procedure consisted of:

- * Assigning concentrations, by compound, to each proposed purge well based on the local groundwater quality data collected in 1986.
- * Weighing the combined influent concentrations to the proposed purge wells according to individual purge well flows.
- * Reducing the weighted results by 50%.



Legend

- Shallow Exploration Wells - 1st Phase
- Shallow Exploration Wells - 2nd Phase
- △ Purge Well
- ⊕ Existing Monitoring Well
- ⋯ DEE Capture Area

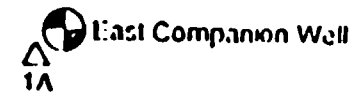
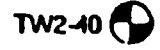
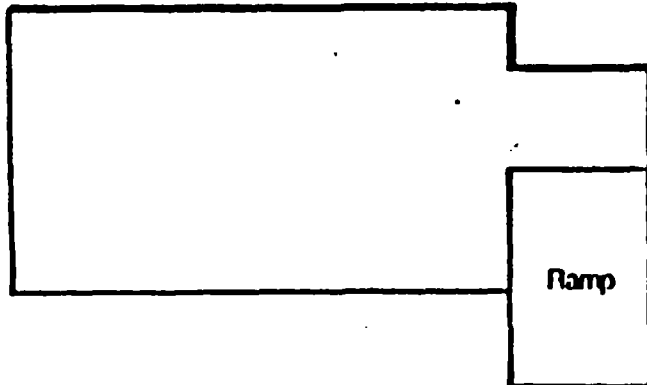
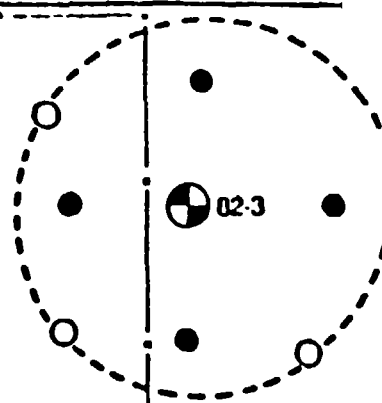
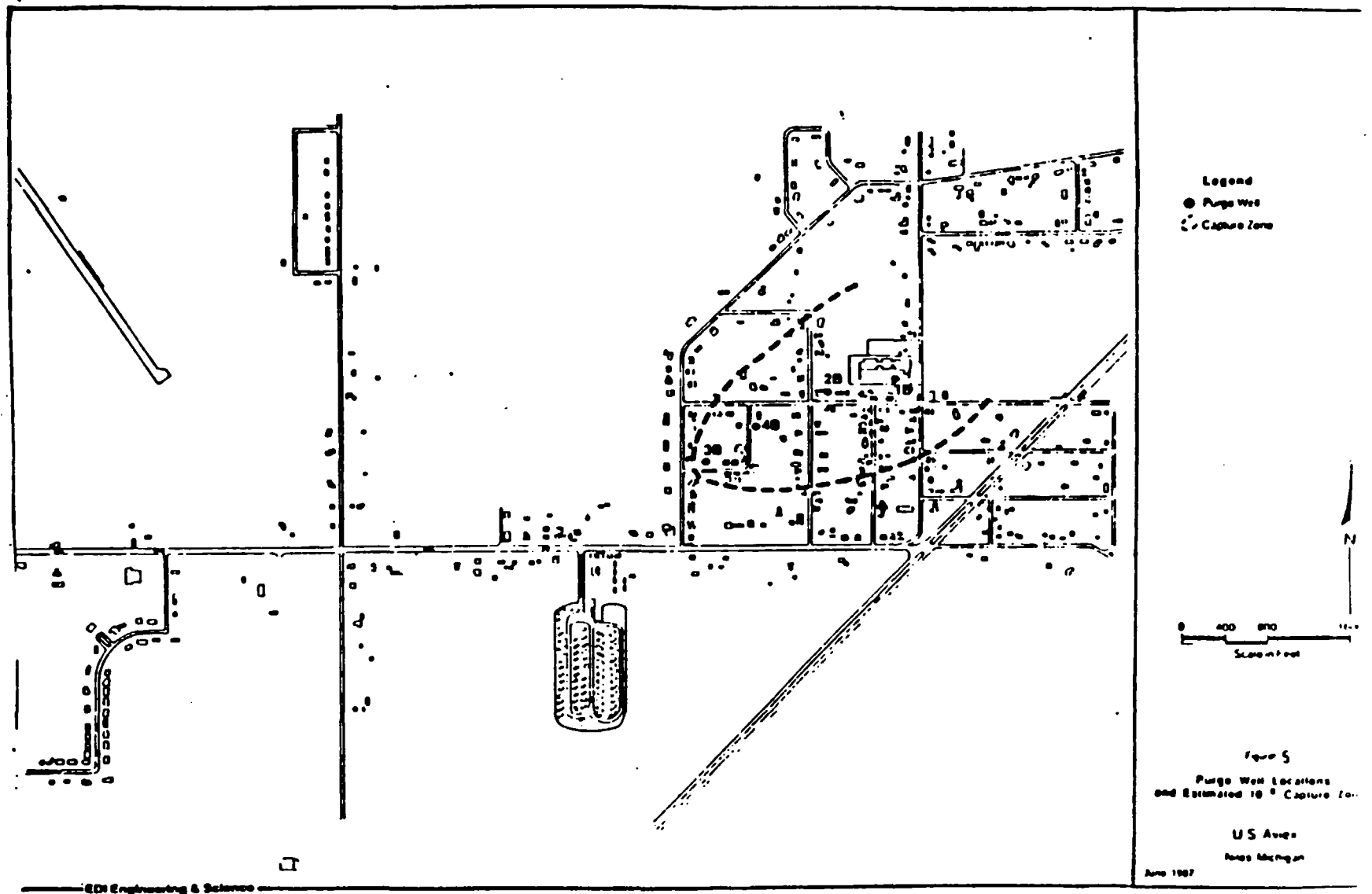


Figure 4
On-Site Purge Well
and DEE Capture Zone

U.S. Aviax



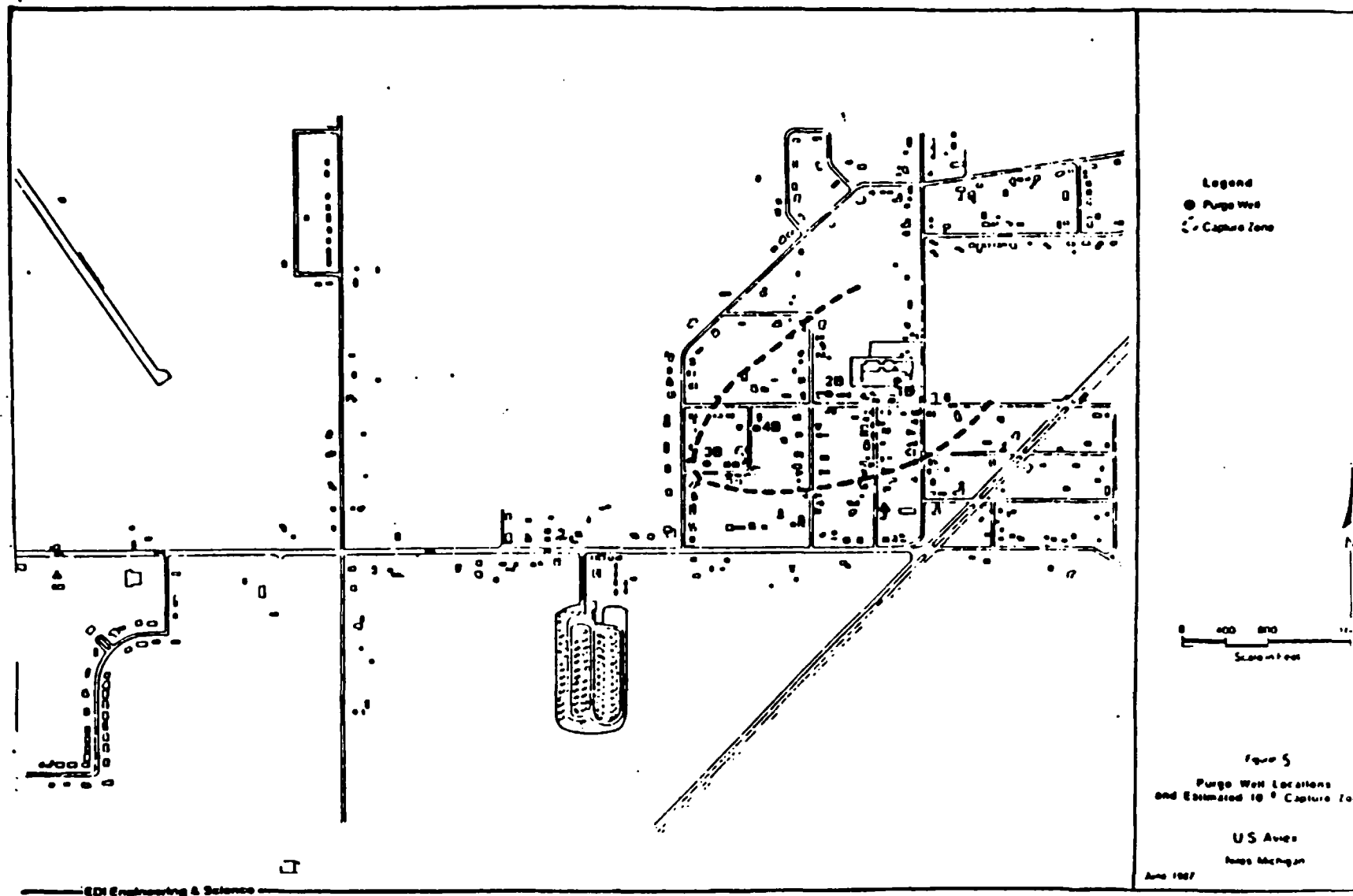


Table 3 includes the data involved in the compound concentration estimates.

C. Groundwater Remediation Goals and Minimum Treated Effluent Discharge Goals

The groundwater cleanup strategy for the U.S. Aviox site is to install and operate purge wells in the plume that will collect and treat on-site and off-site contaminated groundwater found within the U.S. Aviox related 10^{-6} risk plume and in the primary plume, out to its projected limits, (see Figure 7) to comply with existing drinking water standards or EPA or MINR water quality criteria values for human health (see Figure 8). Table 4 lists the clean-up goals for the chemical compounds of concern.

TABLE 4

GROUNDWATER CLEAN-UP GOALS

Compound	Concentration (mg/l)	Risk level	Basis
- Benzene	0.005	4×10^{-6}	MCL
Ethylbenzene	0.680	-	Proposed MCLG
Toluene	2.0	-	Proposed MCLG
Xylene	0.440	-	Proposed MCLG
Chloroform	0.002	1×10^{-6}	Water Quality Criteria for Human Health
1,2-DCA	0.005	1×10^{-5}	MCL
1,1-DCE	0.007	1×10^{-4}	MCL
- TCE	0.005	2×10^{-6}	MCL
PCE	0.00088	1×10^{-6}	Water Quality Criteria for Human Health
Trans-1,2-DCE	0.70	-	Proposed MCL
1,1,1-TCA	0.200	-	MCL
DEE	0.043	-	MINR Rule 57-HLSC*
TCFM	32.0	-	MINR Rule 57-HLSC*
DCFM	3.0	-	MINR Rule 57-HLSC*

* Concentrations proposed but never adopted due to insufficient data.

The proposed groundwater cleanup alternatives(s) will be in operation until the water quality within the plume is remediated to a level that equals or exceeds the individual compound concentrations listed in Table 4. In the process of achieving groundwater cleanup goals, the purged groundwater would be treated to meet minimum concentrations for the various chemicals of concern prior to discharge based on either NPDES permit limits or POTW industrial pretreatment standards. The minimum effluent goals for this site are presented in Table 5. City of Niles POTW pretreatment standards are shown in Table 6.

TABLE 3

Purge Well Compound Concentration Estimation Data/Summary

Purge Well	Compound	Concentration (mg/l)	Data Source	Flows			Remarks
				A Wells In	B Wells In	C Wells	
				Alt B-1/B-2	Alt C-1/C-2 10 ⁻⁶ Risk Plume	Alt C-1/C-2 Primary Plume	
1	1,1,1-TCA	0.8	Existing Purge Wells	100	60	60	Existing combined purge well influent (4/87) x 2 to estimate PW 1 value
	1,2-DCA	0.012	Existing Purge Wells				Existing combined purge well influent (4/87) x 2 to estimate PW 1 value
	1,1-DCA	0.04	Existing Purge Wells				Existing combined purge well influent (4/87) x 2 to estimate PW 1 value
	1,1-DCE	0.015	Existing Purge Wells				Existing combined purge well influent (4/87) x 2 to estimate PW 1 value
	TCFH	0.09	Existing Purge Wells				Existing combined purge well influent (4/87) x 2 to estimate PW 1 value
2(B,C)	DEE	5.6	86-3	-	120	120	86-3 well is the same as on-site monitoring well E-40A
	1,1,1-TCA	2.0	Fig. 21 (R1)				
	1,2-DCA	1.5	86-3				
	1,1-DCA	0.5	Fig. 22 (R1)				
	1,1-DCE	0.1	Fig. 24 (R1)				
	TCFH	0.09	86-2				
3B	1,1,1-TCA	0.1	Fig. 21 (R1)	-	180	-	
	1,2-DCA	0.5	Fig. 23 (R1)				
	DCFH	0.003	Fig. 25 (R1)				
3C	B-T-X (Benzene, toluene, xylene)	0.1	1044 Almagus	-	-	150	Well 3C's location is north of the main plume to effect a combined capture zone with Well 5C. Well 3C may intercept B-T-X based on pre-1986 house well testing by MDPH.
4B	DEE	1.0	Fig. 20 (R1)	-	120	-	
	1,1,1-TCA	0.5	Fig. 21 (R1)				
	1,2-DCA	2.0	Fig. 23 (R1)				
4C	1,1,1-TCA	0.01	Fig. 21 (R1)	-	-	150	
5C	PCE	0.04	Fig. 25 (R1)	-	-	120	

TABLE 3 (CONT'D)

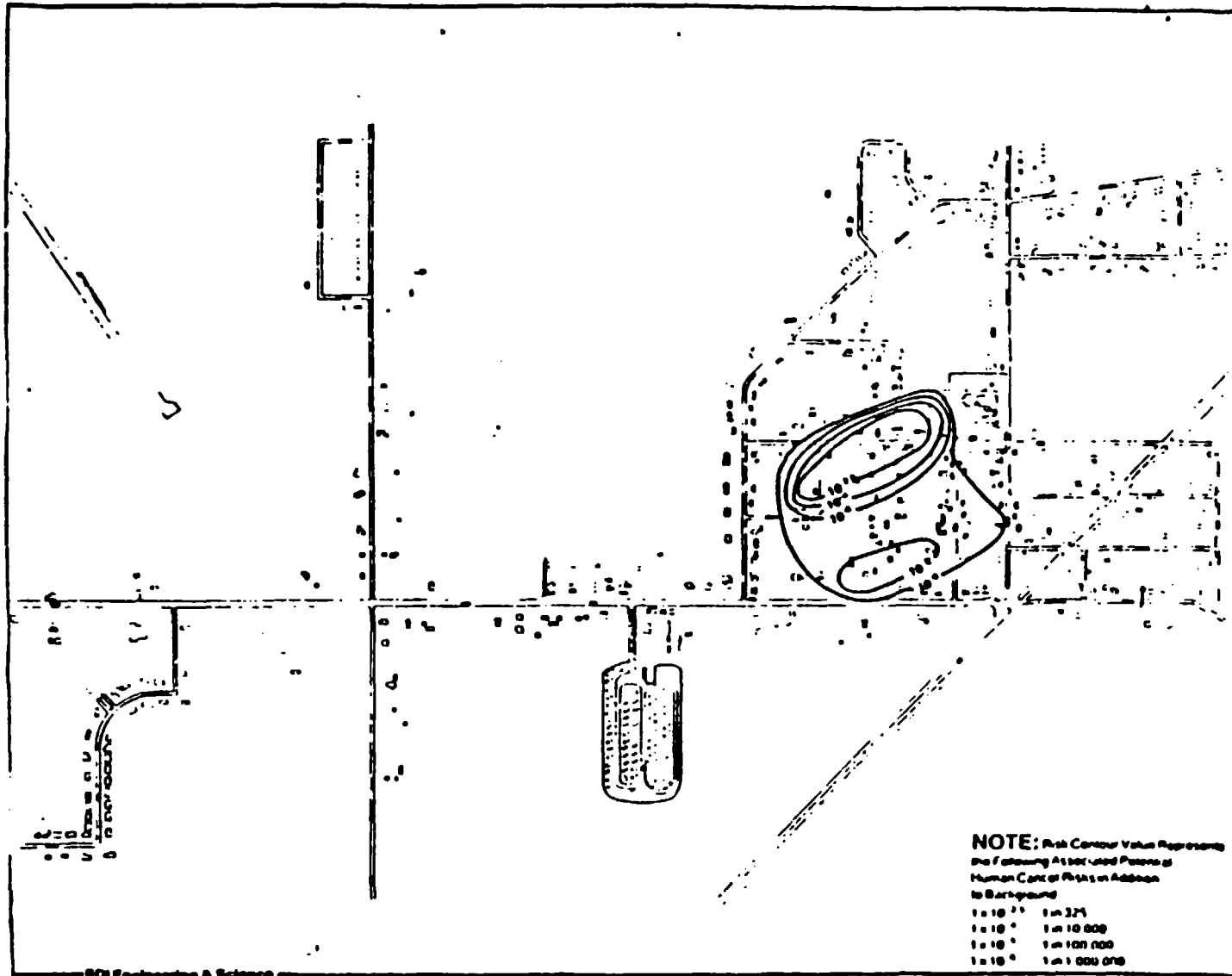
Purge Well	Compound	Concentration (mg/l)	Data Source	Flows			Remarks
				A Wells In Alt B-1/B-2	B Wells In Alt C-1/C-2 10 ⁻⁶ Risk Plume	C Wells Alt C-1/C-2 Primary Plume	
3A	1,2-DCA	1.6	Table 1 (RI)	50	-	-	July 1986 monitoring results E-40, E-40A
	1,1-DCA	1.1	Table 1 (RI)				July 1986 monitoring results E-40, E-40A
	trans-1,2-DCE	0.2	Table 1 (RI)				July 1986 monitoring results E-40, E-40A
	Benzene	0.03	Table 1 (RI)				July 1986 monitoring results E-40, E-40A
Cluster at 82-3	DEE	5.7	Table 1 (RI)	20	20	20	July 1986 monitoring results 82-3
Existing On-site West Purge Well				100	-	-	
	Total Flows (gal./min.)			270	500	620	

Total Concentrations

----- (weighted, mg/l) -----

	A Wells B-1/B-2	B Wells C-1/C-2	C Wells C-1/C-2	A Wells B-1/B-2	B Wells C-1/C-2	C Wells C-1/C-2
DEE	0.42	1.8	1.3	0.21	0.9	0.65
1,1,1-TCA	0.3	0.61	0.47	0.15	0.31	0.24
1,2-DCA	0.3	1.02	0.29	0.15	0.51	0.15
1,1-DCA	0.22	0.13	0.10	0.11	0.065	0.05
1,1-DCE	0.006	0.026	0.023	0.003	0.013	0.012
TCFM	0.033	0.032	0.026	0.017	0.016	0.013
DCFM	-	0.001	-	-	0.0005	-
PCE	-	-	0.008	-	-	0.004
trans-1,2-DCE	0.037	-	-	0.018	-	-
B-1-X	-	-	0.024	-	-	0.012

----- ... x 50% -----



NOTE: Risk Contour Value Represents the Following Assumed Potency of Human Cancer Risks in Addition to Background

1 in 10 ⁻⁶	1 in 32%
1 in 10 ⁻⁵	1 in 10,000
1 in 10 ⁻⁴	1 in 1,000,000
1 in 10 ⁻³	1 in 1,000,000,000

Estimated Contours of Total Human Risk Associated with Maximum Concentrations of Carcinogenic Compounds Detected in Groundwater, 1981.

US Area
New Mexico

November 1981

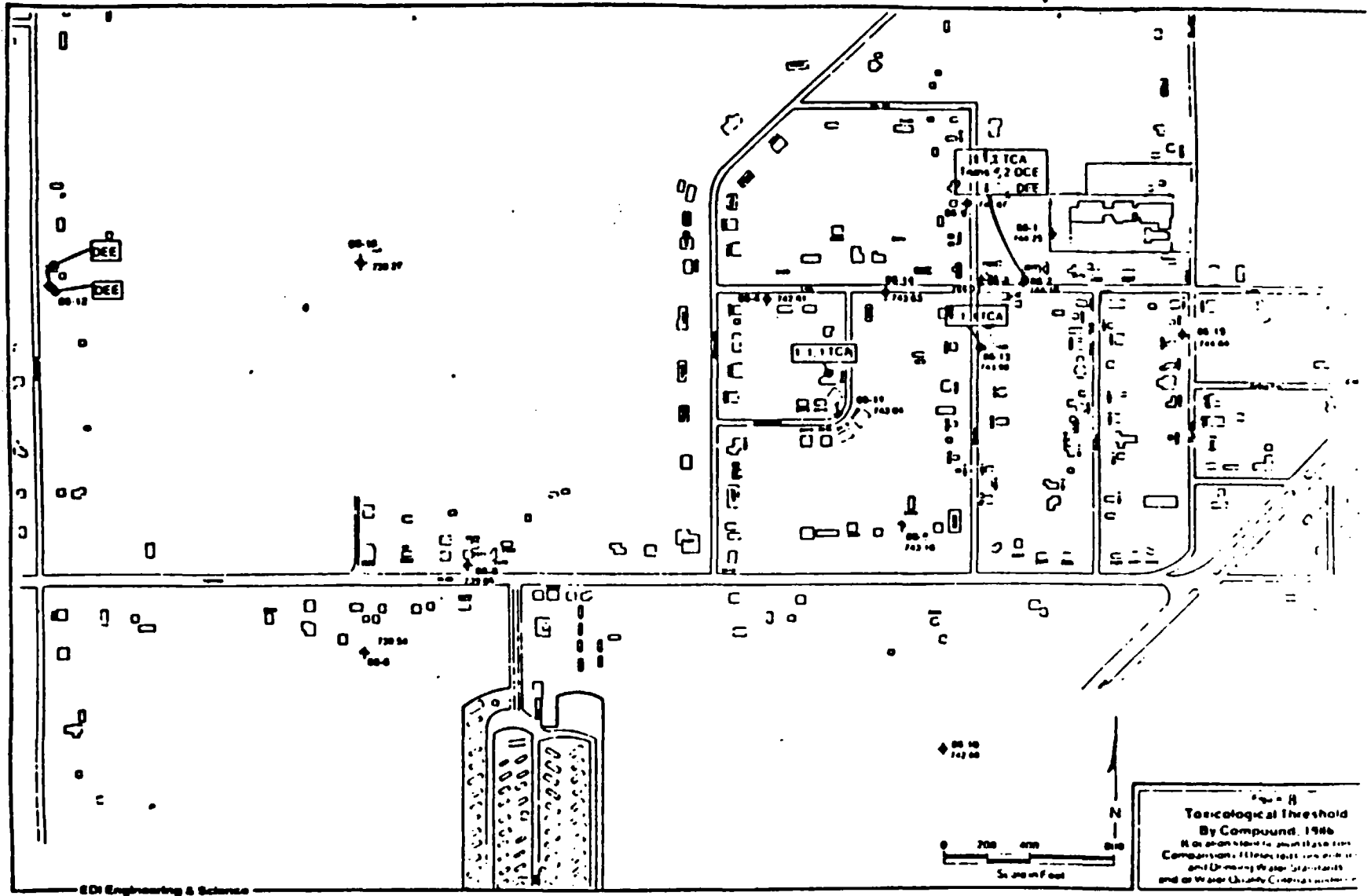


TABLE 5

MINIMUM EFFLUENT GOALS (surface water)

Compound	Concentration (mg/L)	Basis
Benzene	0.051	MINR Rule 57 - ACV
Ethylbenzene	0.062	MINR Rule 57 - ACV
Toluene	0.100	MINR Rule 57 - ACV
Xylene	0.040	MINR Rule 57 - ACV
Chloroform	0.043	MINR Rule 57 - CRV
1,2-DCA	0.560	MINR Rule 57 - CRV
1,1-DCE	0.003	MINR Rule 57 - CRV
TCE	0.094	MINR Rule 57 - ACV
PCE	0.020	MINR Rule 57 - CRV
Trans-1,2-DCE	0.090	MINR Rule 57 - TLSC
1,1,1-TCA	0.120	MINR Rule 57 - ACV
DEE	0.275	Technology Based*
TCFM	0.020	Technology Based*
DCFM	0.020	Technology Based*

AVC - Aquatic Chronic Value

CRV - Cancer Risk Value

TLSC - Terrestrial Life Cycle Safe Concentration *Based on Air Stripping

Diethyl ether (DEE) at 5-6 mg/L persists in a shallow, thin, saturated zone of ground water (5-15 feet thick) at depths ranging from 20 to 40 feet in the area of on-site monitoring well 82-3 (See Figure 4). This saturated zone is underlain with clay. The continued presence of the DEE is due to the poor hydraulic connection between the DEE zone and the part of the aquifer in which the existing purge wells are operating.

Four shallow exploration borings and wells are proposed, see Figure 3, to identify the limits of the DEE zone in the first phase of investigation. The drilling would use hollow-stem auger with split-spoon sampling and field screening of soils at five foot intervals. Two-inch galvanized steel casing with three foot stainless steel well points would be installed, the top of the screen three feet below the water table or with the well point in clay, whichever is more shallow.

These wells will be sampled and analyzed for volatile organics, including DEE, n-propanol, xylene and TCFM. Up to six selected soil samples will be analyzed for the same parameters, if needed to verify the presence of contaminants in the unsaturated zone. Based upon the results of the investigation, up to three additional borings and wells will be installed, sampled, and analyzed as above to establish the spatial limits of on-site DEE contamination.

When the extent of soil and groundwater contamination is determined,

TABLE 4
POLLUTANT DISCHARGE LIMITS
City of Niles
August 1985

		<u>Limit</u>
Arsenic ¹		1.1 mg/l
Cadmium		0.69 mg/l
Chromium (total)		2.77 mg/l
Copper		3.38 mg/l
Lead		0.63 mg/l
Nickel		3.98 mg/l
Silver		0.43 mg/l
Zinc		2.61 mg/l
Cyanide		1.20 mg/l
BOD(5)	Surcharge above 300 mg/l	Upper limit 1,000 mg/l
Suspended Solids	Surcharge above 350 mg/l	Upper limit 1,000 mg/l
Phosphorus	Surcharge above 20 mg/l	Upper limit 100 mg/l
Ammonia	Surcharge above 20 mg/l	Upper limit 100 mg/l
COD		450 mg/l
Chlorine Demand		16.0 mg/l
Fats, Oils, Grease		100 mg/l
pH (must be)		6.0 to 9.5
Heat: Limited to 104°F at plant influent		No discharge above 150°F
Halogenated Hydrocarbons*		
Chloroform		1.9 ug/l
Trichloroethylene		27.0 ug/l
Tetrachloroethylene		8.0 ug/l
1,2-Dichloroethane		9.4 ug/l
Carbon Tetrachloride		4.0 ug/l
BTX		13.0 ug/l
Aromatics*		
Benzene		6.6 ug/l
Chlorobenzene		20.0 ug/l
Dichlorobenzene		400.0 ug/l
3-Tetrachlorobenzene		38.0 ug/l
Pentachlorobenzene		74.0 ug/l
Hexachlorobenzene		7.2 ug/l
Aromatics*		
Benzene		6.6 ug/l
Chlorobenzene		20.0 ug/l
Dichlorobenzene		400.0 ug/l
3-Tetrachlorobenzene		38.0 ug/l
Pentachlorobenzene		74.0 ug/l
Hexachlorobenzene		7.2 ug/l
Phenols*		
Phenol		0.3 mg/l
2-Chlorophenol		0.1 ug/l
4-Chlorophenol		0.1 ug/l
2,4-Dichlorophenol		0.3 ug/l
2,4,6-Trichlorophenol		0.1 ug/l
2,3-Dichlorophenol		0.04 ug/l
2,4-Dichlorophenol		940 ug/l
Phthalate Esters*		

* Data obtained from U.S. EPA Treatability Manual, Volume 1,
EPA-600/2-82-001a, September 1981.

If a particular remedial action alternative met one or more of the above criteria then it was considered not appropriate for the site. A summary of remedial alternative screening is presented in Table 7. The following alternatives were deemed not appropriate for use at the U.S. Aviox site.

On-Site Soils

No Action; No Treatment of On-site Soils

Not consistent with the clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, or provide a permanent solution. Also is not consistent with remedial action objectives established for this site.

Control leachate by removal of soils and off-site disposal in an Act 64 landfill.

Not consistent with the clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, reduce its volume, or provide a permanent solution in keeping with cradle-to-grave responsibilities.

On-Site Groundwater

No Action A: No Treatment of On-site Groundwater.

Not consistent with the clean-up criteria established by SARA of 1986: Does not destroy or detoxify the waste or reduce its mobility or volume. Also not consistent with the remedial action objectives established for this site.

No Action B: Continue with Existing Purge and Treatment System

Capture and treatment of on-site groundwater will continue to be needed but in a combined configuration with off-site purge wells that could result in elimination of an existing purge well.

Collection with On-site Purge Well and Disposal by On-site Deep-well Injection: Excessive relative cost (approximately \$2,000,000 to permit and install only). Not consistent with clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, or reduce its volume and excessive time (up to two years) is required to obtain approval, if at all, due to permit requirements.

Collection with On-site Purge Well and Disposal by Off-site Deep-well Injection: Excessive relative cost (approximately \$2,000,000 to permit and install only). Not consistent with clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste or reduce its volume, excessive time required to obtain approval, if at all, (up to two years) due to permit requirements. Capture and treatment of on-site groundwater will continue to be needed but in a combined configuration with off-site purge wells that could result in elimination of an existing purge well or

TABLE 7

SUMMARY OF REMEDIAL ALTERNATIVE SCREENING

Remedial Alternative	Site Conditions	Tech	SARA	Implementation Timeframe	Relative Cost	Environmental Consideration	Screen Decision
On-Site Source Control-Soils							
No Action	-	-	Not Consistent Not Permanent Selection	-	-	Inconsistent with remedial action obj.	E
In-situ-Aeration	Amenable	Proven	Consistent	Acceptable	Acceptable	Good	R
In-situ-Flushing	Amenable	Proven	Consistent	Acceptable	Acceptable	Good	R
Capping	With Modif. Amenable	Proven	Not Consistent Not Permanent Sol.	Acceptable	High	Does not eliminate source	R
Removal/Off-Site Disposal							
- Aeration/641 Landfill	Amenable	Proven	Not Consistent	Moderate	High	Uses up Landfill Space, Release of VOC's	R
- Incineration/641 Landfill	Amenable	Proven	Not Fully Consistent	Moderate	High	Uses up Landfill Space	R
I.A.3.c Removal/Off-Site Landfill	Amenable	Proven	Not Consistent	Moderate	High	Uses up Landfill Space, Release VOC's	E
On-Site Source Control-Groundwater							
No Action A	-	-	Not Consistent Not Permanent	-	-	Inconsistent with Remedial Action Obj.	E
No Action B	Amenable	Proven	Consistent	Long-term	Acceptable	Need to Expand Existing System to Comply	E
On-Site Collection, Treatment with Off-Site Disposal	Amenable	Proven	Consistent	Long-term	POTW High Drain Acceptable	Good	R
On-Site Collection with Off-Site Treatment/ Disposal	Amenable	Proven	Consistent	Long-term	POTW High Drain Acceptable	May not meet POTW Standards	R
On-Site Collection, Disposal by On-Site Deep Well Injection	Available Underground Formation Questionable	Proven with Reservations	Not Consistent	Moderate-term	Excessive	Questionable	E
On-Site Collection, Disposal by Off-Site Deep Well Injection	Available Underground Formation Questionable	Proven with Reservations	Not Consistent	Moderate-term	Excessive	Questionable	E

Table 7 (Cont'd)

Remedial Alternative	Site Conditions	Tech	SARA	Implementation Timeframe	Relative Cost	Environmental Consideration	Screen Decision
Collect, Store, and Disposal Untreated by:							
Off-Site Deep Well Injection	Available Underground Formation Questionable	Proven with reservations	Not Consistent	Long-term	Excessive	Questionable	E
On-Site Deep Well Injection	Available Underground Formation Questionable	Proven with reservations	Not Consistent	Long-term	Excessive	Questionable	E
Collect, Temporary Storage, Discharge Untreated Groundwater to Surface Drain	Amenable	Proven, storage not required	Not consistent	Long-term	Acceptable	Would not meet NPDES	E
Collect, Temporary Storage, Discharge to Niles POTW	Amenable	Proven, storage not required	Consistent	Long-term	High	POTW may not accept	R
Collect, Transport Back to Site, Store, Treat/Disposal							
Air stripping with Dame-Huntly discharge	Amenable	Proven, storage not required	Consistent	Long-term	Acceptable	Good	R
Air stripping with Niles POTW discharge	Amenable	Proven, storage not required	Consistent	Long-term	High	POTW may not accept	R
Steam stripping with Dame-Huntly discharge	Amenable	Proven, storage not required	Consistent	Long-term	Acceptable	Good	R
Steam stripping with Niles POTW discharge	Amenable	Proven, storage not required	Consistent	Long-term	High	POTW may not accept	R
Carbon adsorption with Dame-Huntly discharge	Amenable	Proven, storage not required	Consistent	Long-term	Acceptable	Good	R
Carbon adsorption with Niles POTW discharge	Amenable	Proven, storage not required	Consistent	Long-term	High	POTW may not accept	R

Table 7 (Cont'd)

Remedial Alternative	Site Conditions	Tech	SARA	Implementation Timeframe	Relative Cost	Environmental Consideration	Screen Decision
Management Migration Groundwater (off-site/on-site)							
No Action	-	-	No permanent solution Not consistent	-	-	Inconsistent with Remedial Action Obj.	E
Monitor Only	Amenable	No Clean-up	Not a permanent solution, not consistent	Long-term	Acceptable	Inconsistent with remedial action obj.	E
Collect, store, transport to treat system via pipeline with off-site treat/disposal							
Air stripping with Bame-Huntly discharge	Amenable	Proven, but impractical; storage not required	Consistent	Long-term	Acceptable	Siting	E
Air stripping with Niles POTW discharge	Amenable	Proven, but impractical; storage not required	Consistent	Long-term	High	POTW may not accept	E
Steam stripping with Bame-Huntly discharge	Amenable	Proven, but impractical; storage not required	Consistent	Long-term	Acceptable	Siting	E
Steam stripping with Niles POTW discharge	Amenable	Proven, but impractical; storage not required	Consistent	Long-term	High	POTW may not accept	E
Carbon adsorption with Bame-Huntly discharge	Amenable	Proven, but impractical; storage not required	Consistent	Long-term	Acceptable	Siting	E
Carbon adsorption with Niles POTW discharge	Amenable	Proven, but impractical; storage not required	Consistent	Long-term	High	POTW may not accept	E

Table 2 (Cont'd)

Page 4 of 4

Remedial Alternative	Site Conditions			Implementation Timeframe	Relative Cost	Environmental Consideration	Screen Decision
Alternative Drinking Water Supplies							
Howard Twp. Municipal Well Syst.	-	Pro	-	Short-term Permanent	>\$1,000,000	Meets remedial action obj.	R
Connect to Miles Municipal Syst.	-	Prove	-	Short-term Permanent	>\$1,000,000	Meets remedial action obj.	R
Install deepened Municipal Syst.	-	Proven	-	Short-term Permanent	<\$1,000,000	Must confirm deep aquifer protected	R
Individual Domestic Well Treatment	-	Proven but limited	-	Short-term	<\$1,000,000	Marginal	R
Provide Bottled Water to Affected Residence	-	Proven but not desirable	-	Short-term	Excessive Relative Cost	Socially unacceptable	E
Relocate Affected Residence	-	Not Feasible	No realistic solution, not consistent	Long-term Permanent	Excessive Relative Cost	Socially unacceptable	E

E = Eliminate from further consideration.

R = Retain for detailed evaluation.

increasing the flow to the existing air stripper.

On-Site and Off-Site Groundwater

No Action A: No treatment of on-site or off-site groundwater.

Not consistent with clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, reduce its volume or mobility, or provide a permanent solution. Also not consistent with the remedial action objectives established for this site.

Monitoring Only: Not consistent with clean-up criteria established by SARA of 1986: Does not destroy or detoxify waste, reduce its volume or mobility, or provide a permanent solution. Also not consistent with the remedial action objectives established for this site.

Collection with Purge Well(s) and Disposal by Off-site Deep Well injection: Excessive relative cost (approximately \$2,000,000 to permit and install only). Not consistent with clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, or reduce its volume, excessive time required to obtain approval, if at all, (up to two years due to permit requirements).

Collection with Purge Well(s), and Discharge of Untreated Groundwater to a Surface Drain: Not consistent with clean-up criteria established by SARA of 1986. Does not destroy the waste, reduce its volume, or provide a permanent solution and is not consistent with remedial action objectives established for this site.

Provide Bottled Water to Affected Residents: Not consistent with clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, reduce its volume or mobility, or provide a permanent solution. The cost is excessive when compared to other alternatives.

Relocate Affected Residents: Not consistent with clean-up criteria established by SARA of 1986. Does not destroy or detoxify the waste, reduce its volume or mobility, or provide a permanent solution. The cost is excessive when compared to other alternatives.

VI. Description of Alternatives

Acceptable alternatives to address the remedial objectives at the U.S. Aviex site have two goals or objectives. First, they must control the on-site source of contamination, and second, they must control and treat the on-site and off-site contaminant plumes. The on-site control alternatives are directed toward sub-surface soil contamination through in-situ treatment or through leachate control by capping or excavation and incineration. An outline and discussion of these alternatives follows.

A. On-site Source Control - Soils

1. In-situ treatment

a. Aeration

- b. Soil flushing
 - 2. Leachate Control
 - a. Capping
 - b. Excavate and Incinerate
- B. On-site Source Control - Groundwater
 - 1. On-site Collection and Treatment, Off-site Disposal
 - a. Air Stripping
 - b. Steam Stripping
 - c. Carbon Adsorption
- C. Management of Migration - On-site and Off-site Groundwater
 - 1. On-site and Off-site Collection, On-site Treatment with Off-site Disposal

Alternative A.1.a. Aeration

An aeration system pulls air through the unsaturated soils, extracting volatile organic compounds (VOCs) from the pore spaces in the soil. The aeration equipment includes a system of air intake wells, air withdrawal wells, piping and a blower. Air intake wells are placed around the perimeter of the contaminated areas. Air withdrawal wells are placed in the center of the contaminated areas. A blower draws atmospheric air into the intake wells, through the soil pore space, and into the withdrawal wells. A vapor carbon absorber in line with the blower would remove the VOCs from the air before discharging into the atmosphere. A plastic membrane placed over the contaminated area ensures that air removed from the withdrawal well is drawn through the pore space and not from the surface near the withdrawal wells.

Alternative A.1.b. Soil Flushing

A soil flushing system removes VOCs from the contaminated soils by allowing water (possibly mixed with surfactants) to flow through the soil pore space. The contaminated water is then removed from the ground by the on-site purge wells and treated in the selected treatment system. Soil flushing takes place to some extent naturally due to the infiltration of rainwater in the cracks of the existing pavement that covers the contaminated areas. Soil flushing would be accomplished by an underground system of perforated pipelines below grade in the areas of contamination. The water supply could be obtained from an on-site well drilled to the uncontaminated aquifer or by connecting to the Niles Municipal Water System. The well would be constructed in such a way as to prevent contamination of the lower aquifer. Water (with a surfactant supplied through a chemical feed pump if required) would be continually applied to the subsurface soils.

Leachate production from the contaminated soils can be controlled by capping the surface to minimize surface water infiltration or eliminated by removing the soils.

Alternative A.2.a. Capping

The capping alternative reduces the permeability of the site surface,

minimizing infiltration. The multi-layered cap would consist of four feet of clay, a twenty mil PVC synthetic membrane, eighteen inches of sand, and six inches of top soil. The top of the cap would be seeded, mulched and fertilized. Post closure monitoring would be required to ensure continued effectiveness of the cap. The capping alternative would require excavation of approximately six feet of soil prior to installation because of site limitations. The excavated soil would be incinerated at a RCRA compliant facility.

Alternative A.2.b. Excavation and Incineration

Excavation of the contaminated soils is complicated by the depth of the soils to be removed and their proximity to buildings. Sheet piling will be necessary to protect the adjacent buildings during excavation. Since the required excavation depth, twenty feet, exceeds the depth that cantilever sheet piling may be used, a braced cofferdam system on all sides of the excavations will be necessary. Excavated soils would be disposed of at a RCRA approved landfill after treatment by incineration. The excavated area would be backfilled with off-site granular soils and the asphalt and concrete surfaces restored as required.

Alternative B.1. On-site Collection and Treatment with Off-site Disposal

On-site treatment can be accomplished with air stripping, steam stripping, or carbon adsorption. Off-site disposal options include discharge to the St. Joseph River via the Bame-Huntly Drain or to the Niles wastewater treatment plant POTW. All three treatment options require removal of the groundwater with on-site purge wells and from a separate well point purge area on-site designed specifically to capture the zone of DEE found and isolated on-site.

The following values were used as the basis of the design and cost analysis for each treatment system:

Flow Rate	270 gal/min
Temperature	55°F
Contaminant Loading	
1,2-DCA	150 ppb
1,1,1-TCA	150 ppb
1,1-DCE	3 ppb
Trans-1,2-DCE	18 ppb
PCE	<1 ppb
B-T-X	<1 ppb
(Benzene-Toluene-Xylene)	
DEE	210 ppb
TCFM	17 ppb
DCFM	<1 ppb

Effluent concentrations from each treatment system treating the 10^{-6} risk plume groundwater are estimated below:

Compound	Air Stripping	Steam Stripping	Carbon Adsorption
----------	------------------	--------------------	----------------------

1,2-DCA	50 ppb	<1 ppb	<1 ppb
1,1,1-TCA	<1 ppb	<1 ppb	<1 ppb
1,1-DCE	<1 ppb	<1 ppb	<1 ppb
Trans-1,2-DCE	<1 ppb	<1 ppb	<1 ppb
PCE	<1 ppb	<1 ppb	<1 ppb
B-T-X	<1 ppb	<1 ppb	<1 ppb
DEE	70 ppb	<1 ppb	<1 ppb
TCFM	<1 ppb	<1 ppb	<1 ppb
DCFM	<1 ppb	<1 ppb	<1 ppb

This alternative will prevent migration of the contaminants off-site but it will not treat groundwater that has already migrated off-site. Groundwater monitoring will be used to assure that no further off-site migration occurs.

Alternative B.1.a. Air Stripping

Currently, the on-site collection and treatment of the groundwater is accomplished by an air stripping tower that discharges to the Bame-Huntley Drain. The air stripping process mixes large volumes of air with water in a packed column to promote the transfer of VOCs into the air. Groundwater from two purge wells is pumped to the top of a packed tower where it is distributed over a bed of plastic media. Air introduced at the bottom of the tower passes countercurrent to the groundwater, stripping the contaminants from the groundwater. The air containing the VOCs is currently released to the atmosphere because it meets Air Quality Standards.

To completely eliminate migration of the groundwater off-site, another purge well will be added to the existing two well system. Due to increased contaminant loading, it may also be necessary to add air controls to the emissions from the air stripping tower. Because of the increased groundwater purge rate, a larger air stripping column will be required.

Alternative B.1.b. Steam Stripping

A steam stripping treatment system is similar to an air stripping treatment system except that steam is the stripping media rather than air. The added heat increases the volatility of the contaminants. Additional equipment is also required to condense the steam.

Alternative B.1.c. Carbon Adsorption

A carbon adsorption treatment system involves filters the contaminated groundwater through two tanks filled with activated carbon. The contaminants absorb onto the carbon surface and are removed from the groundwater. Periodically the carbon must be replaced or regenerated.

In addition to the purge and on-site treatment and off-site disposal of the groundwater beneath the site, this alternative, as well as all of the pump and treat alternatives, must treat the leachate produced from the zone of isolated DEE contamination. This soil is located above a semi-isolated

shallow groundwater zone underlain by clay.

Effluent from any of the treatment options could be discharged to the Bame-Huntley Drain or to the Niles wastewater treatment plant. Discharge to the Bame-Huntley Drain would require a National Pollution Discharge Elimination System (NPDES) permit. U.S. Aviex has a NPDES permit for its present discharge. This permit would have to be modified for the increased flow and contaminant loading. To discharge to the Niles wastewater treatment plant, a pipeline must be constructed to the nearest sewer with adequate capacity for the flow. This sewer is located near the intersection of 17th and Oak Streets, approximately 2.5 miles from the U.S. Aviex site. Currently the wastewater treatment plant has sufficient capacity to accept the wastewater flow.

Flow rate and influent concentration would be identical to that of the on-site treatment alternatives. Off-site treatment would be at the Niles wastewater treatment plant. A pipeline would be constructed to discharge the purged groundwater to the plant. An existing sanitary sewer has adequate capacity to handle this flow.

Alternative C.1. Management of Migration of Groundwater. On-site Treatment with Off-site Disposal

This alternative may be divided into two options. The difference between the options is the extent of the contaminant plume capture. In the first option, the wells would be placed off-site to capture the 10^{-6} risk plume. In addition, one of the on-site wells would control the on-site leachate (4 wells total). A total volume of 500 gal/min of groundwater would be treated and disposed of. The second option would also use one on-site well but would involve four wells to control off-site leachate migration (5 wells total). These wells, placed at different locations than the wells in the first option, would collect groundwater at a rate of 620 gal/min, capturing the migrating flow from the primary contaminant plume. Each option also requires the wellpoint collection of 20 gal/min of on-site groundwater contaminated with DEE.

The following values were used as the basis for the design and cost analysis for each treatment system for the 10^{-6} risk plume:

Flow Rate	500 gal/min
Temperature	55°F
Contaminant Loading	
1,2-DCA	510 ppb
1,1,1-TCA	240 ppb
1,1-DCE	13 ppb
Trans-1,2-DCE	<1 ppb
PCE	<1 ppb
B-T-X	<1 ppb
DEE	900 ppb
TCFM	16 ppb

B-T-X	<1 ppb
DEE	900 ppb
TCFM	16 ppb
DCFM	<1 ppb

Effluent concentrations from each treatment system treating the 10^{-6} risk plume groundwater are estimated to be:

Compound	Air Stripping	Steam Stripping	Carbon Adsorption
1,2-DCA	160 ppb	<1 ppb	<1 ppb
1,1,1-TCA	<1 ppb	<1 ppb	<1 ppb
1,1-DCE	<1 ppb	<1 ppb	<1 ppb
Trans-1,2-DCE	<1 ppb	<1 ppb	<1 ppb
PCE	<1 ppb	<1 ppb	<1 ppb
B-T-X	<1 ppb	<1 ppb	<1 ppb
DEE	275 ppb	<1 ppb	<1 ppb
TCFM	<1 ppb	<1 ppb	<1 ppb
DCFM	<1 ppb	<1 ppb	<1 ppb

The following values were used as the basis for the design and cost analysis for each treatment system for the primary plume:

Flow Rate Total	620 gal/min
Temperature	55°F
Contaminant Loading	
1,2-DCA	150 ppb
1,1,1-TCA	240 ppb
1,1-DCE	12 ppb
Trans-1,2-DCE	<1 ppb
PCE	4 ppb
B-T-X	10 ppb
DEE	650 ppb
TCFM	13 ppb
DCFM	<1 ppb

Effluent concentrations for each treatment system treating the primary plume groundwater are estimated to be:

Compound	Air Stripping	Steam Stripping	Carbon Adsorption
1,2-DCA	50 ppb	<1 ppb	<1 ppb
1,1,1-TCA	<1 ppb	<1 ppb	<1 ppb
1,1-DCE	<1 ppb	<1 ppb	<1 ppb
Trans-1,2-DCE	<1 ppb	<1 ppb	<1 ppb
PCE	<1 ppb	<1 ppb	<1 ppb
B-T-X	<1 ppb	<1 ppb	<1 ppb
DEE	200 ppb	<1 ppb	<1 ppb
TCFM	<1 ppb	<1 ppb	<1 ppb
DCFM	<1 ppb	<1 ppb	<1 ppb

Each of these alternatives has several options. On-site treatment can be accomplished by air stripping, steam stripping, or carbon adsorption. Off-site disposal options include discharge to the St. Joseph River via the Bame-Huntley Drain or discharge to the Niles wastewater treatment plant as described earlier. In either collection alternative, 10^{-6} risk plume control or primary plume control, a groundwater monitoring program

will be implemented to ensure that the desired degree of plume capture is occurring.

An alternative with off-site treatment and disposal is also considered for the combined collection alternative. The collection is the same as above but flushing fluids discharge to the Niles POTW.

VII. Summary of the Comparative Analysis of Alternatives

The alternatives described above were found, in the FS screening, to merit detailed analysis. The analysis of these alternatives measured against the appropriate criteria follows.

1. Overall Protection of Human Health and the Environment addresses whether or not a remedy provides adequate protection, and describes how risks are eliminated, reduced or controlled through treatment engineering controls, or institutional controls.

The soil is treated to eliminate the risk from exposure through ingestion or inhalation of contaminated soil. Alternatives for treatment of the contaminated soil through in-situ treatment reduce the risk to the public health and the environment.

The capping alternative prevents exposure to contaminated soil and reduces the production of contaminated groundwater (leachate). The continued effectiveness of this alternative is dependent upon land use restrictions and maintaining the integrity of the cap.

The on-site treatment of contaminated groundwater alternatives reduce the risk to human health and the environment by eliminating the source of exposure as well as eliminating potential contamination of off-site groundwater. Restricting the use of on-site groundwater during the period of treatment eliminates the risk of exposure through ingestion.

Management of the migrating off-site groundwater contaminants through collection and treatment protects the public health and the environment. During the period of treatment, groundwater use will be restricted through institutional controls.

2. Compliance with ARARs addresses whether or not an alternative will meet all of the applicable or relevant and appropriate requirements of other environmental statutes and/or provide grounds for invoking a waiver. ARARs are summarized in Table 8 for each alternative.

The Clean Air Act, 40 CFR 50 and 52, which implements and sets rules for regional air pollution control and establishes Ambient Air Quality Standards, is applicable to discharges of toxic substances to the atmosphere during waste handling or treatment. This Act is administered by Michigan Department of Natural Resources (MDNR) Air Quality Division under Michigan Act 348. The Air Quality Division is authorized to ensure that a person does not cause or permit the emissions of an air contaminant in quantities that will cause injurious effects to human health or safety,

animal life, plant life of significant economic value, or property "or unreasonable interference with the comfortable enjoyment of life and property." The Air Quality Division requires appropriate air monitoring of emissions and compliance with Federal PSD regulations, Federal NSPS, and NESHAP standards.

These standards will be met during treatment of soil and groundwater through the use of discharge control systems designed to meet the requirements of Rules 201 and 203 of Michigan Act 348. The capping and excavation and incineration alternatives will not meet Michigan Act 348 requirements and standards. However, only air monitoring to demonstrate compliance with Rule 901 of Act 348 has been required for installation of slurry walls.

The Clean Water Act 40 CFR regulates point discharges to navigable waters. This Act is administered by the MDNR under Michigan Act 245 and establishes surface water quality standards "to protect the public health and welfare, to enhance and maintain the quality of the water, to protect the State's natural resources." The MDNR oversees point discharge standards as promulgated by the Federal NPDES program under this Act. The groundwater treatment alternatives (air stripping, steam stripping, and carbon adsorption) will provide levels of efficiency meeting NPDES permit standards. Part 9 of Michigan Act 245 requires reporting of critical materials. At this site the defined critical materials include benzene, 1,2-DCA, Trichloroethylene, and Tetrachloroethylene. The alternatives will meet the requirements for an effluent discharge permit and the terms and conditions of the permits effluent standards and limitations.

The State has identified Michigan Act 245, Part 22, which regulates groundwater quality, as an ARAR. The purpose of the groundwater quality regulations is "to protect the public health and welfare and to maintain the quality of groundwater in all usable aquifers for individual, public, industrial and agricultural water supplies and provide for the nondegradation of groundwater quality in usable aquifers" by preventing discharge into the groundwater. Michigan Act 245 is not applicable since no discharges into the groundwater are proposed. Michigan Act 245 is not relevant or appropriate to establishing clean-up levels or areas of attainment at the site since no standards are provided. However, the groundwater treatment alternative proposes treatment to the 10^{-6} risk levels which will reduce contaminant concentration levels below the MCLs.

Installation of all wells in the aeration, flushing, and groundwater treatment alternatives will conform to Michigan Act 315 and the administrative rules which regulate construction, operation, and abandonment of such wells.

Michigan Act 98 regulates supervision and control over sewage systems including planning, construction, operation and maintenance. The groundwater treatment alternatives (air stripping, steam stripping and carbon adsorption) will meet the requirements of this Act.

Michigan Act 346 regulates inland lakes and streams. Act 346 requires

TABLE 8
ARARs FOR THE U.S. AVIEX SITE

Regulation	Requirement	Regulated Action or Condition	Alternative				
			A.1.a.	A.1.b.	A.2.a.	B.and C.	A.2.b.
FEDERAL							
Clean Air Act, Section 101	Implementation of regional air pollution control programs.	Emissions during excavation, treatment of soil and groundwater	X	X	X	X	X
40 CFR 52	Rules for implementation of regional air quality plan.	Emissions during excavation, treatment of soil and groundwater	X	X	X	X	X
40 CFR 50	Air quality standards.	Emissions during excavation, treatment of soil and groundwater	X	X	X	X	X
Clean Water Act, Section 208	Actions consistent with water quality management program.	Discharges to Bame-Huntley Drain		X		X	X
40 CFR 122	NPDES permit regulations.	Discharges to Bame-Huntley Drain		X		X	X
40 CFR 125	Criteria for NPDES permit.	Discharges to Bame-Huntley Drain		X		X	X
40 CFR 131	Enforcement jurisdiction over Discharges granted to States.	Discharges to Bame-Huntley Drain		X		X	X
40 CFR 136	Adherence to sampling Procedures.	Discharges to Bame-Huntley Drain		X		X	X
40 CFR 261	Definition and identification of hazards waste.	Excavated materials and groundwater treated residuals.			X	X	X
40 CFR 262	Regulations for generation of hazardous waste	Groundwater treatment residuals hauled off-site.				X	X
40 CFR 263	Regulations for transport of hazardous waste	Groundwater treatment residuals hauled off-site.				X	X

Table B (Page 2 of 3)

<u>Regulation</u>	<u>Requirement</u>	<u>Regulated Action or Condition</u>	<u>Alternative</u>				
			<u>A.1.a.</u>	<u>A.1.b.</u>	<u>A.2.a.</u>	<u>D.and C.</u>	<u>A.2.b.</u>
40 CFR 264 Subpart X	Standards for environmental performance of miscellaneous treatment units.	Physical/chemical treatment.					X
40 CFR 268 Subparts C and D	Land-banned waste materials.	Would apply to certain solvent-wastes.					X
40 CFR 403	Pretreatment standards for discharge to local POTW.	Discharges to POTW after installation of extraction wells but before installation of on-site groundwater treatment plant.				X	
40 CFR 141	Established maximum contaminant levels (MCLs) for drinking water quality.	Groundwater at the site.		X		X	
Executive Order 12372 40 CFR 29	Requires state and local coordination and review of EPA-assisted projects.	U.S. Avtex Site is a CERCLA site.	X	X	X	X	X
<u>State</u>							
ACT 348	Regulates air emissions and requires monitoring of air.	In-situ soil treatment RCRA, Cap	X	X	X	X	X
ACT 315	Regulates permitting, construction, and abandonment of wells.	Monitoring wells on-site and near site. Groundwater extraction wells.	X	X	X	X	X
Act 64 Part 2 and 3	Requirements for identification, recordkeeping, and management of hazardous waste.	Generated groundwater treatment residuals and excavated materials.			X	X	X

Table 8 (Page 3 of 3)

<u>Regulation</u>	<u>Requirement</u>	<u>Regulated Action or Condition</u>	<u>Alternative</u>				
			<u>A.2.a.</u>	<u>A.1.b.</u>	<u>A.2.a</u>	<u>B. and C.</u>	<u>A.2.b.</u>
ACT 64, Part 5	Regulations for construction and licensing treatment facility.	Groundwater treatment		X			X
ACT 64, Part 6	Specifies technical standards for locating operating and closing a hazardous waste facility.	Groundwater treatment		X	X	X	X
ACT 245 Part 9	Rules for reporting discharges to state waters.	Discharges from groundwater treatment.		X		X	
ACT 245, Part 21	Rules for NPDES permit.	Discharges from groundwater treatment.		X		X	
ACT 98	Supervision and control of sewage systems	Construction and operation of groundwater treatment.				X	
ACT 346 Section 3	Permit for construction in "bottom land."	Buried pipeline for groundwater water collection and discharge.				X	
ACT 368	Authority to safeguard public health, determine imminent danger.	Any activity	X	X	X	X	X

compliance with permit regulations to "dredge or fill bottomland" or to "construct, enlarge, extend, remove, or place a structure on bottomland."

Act 346 specifies the technical information and criteria required for evaluating and reviewing the proposed activities. The construction of surface water discharge piping for disposal of treated groundwater will meet these requirements.

40 CFR 260-264 RCRA Subtitle C, regulates the generation, transportation, storage, treatment and disposal of hazardous wastes generated in the course of remedial action. The program is administered under Michigan Act 64. The Administrative rules requiring identification of hazardous wastes and compliance with management requirements will be met for carbon adsorption from the groundwater treatment alternative and for the emission control system of the air stripping or steam stripping alternatives. Requirements for construction and operation of a hazardous waste treatment facility in Part 5 and Part 6 of Michigan Act 64 will be met by the treatment alternatives.

The Safe Drinking Water Act, 40 CFR 141, establishes Maximum Contaminant Levels (MCLs) for drinking water quality. The treatment alternatives for on-site and off-site groundwater will meet the MCLs within the area of attainment based upon clean up levels of 10^{-6} total risk. Risks associated with the MCLs range from 10^{-4} to 10^{-6} .

3. Long-term Effectiveness and Permanence refers to the ability of a remedy to maintain reliable protection of human health and the environment over time once cleanup goals have been met.

Both the soil flushing and the aeration alternative for on site soil source control provide long-term effectiveness and permanence through treatment. Under the soil flushing alternative levels of contamination are reduced until the generated leachate (flushing fluids) no longer acts as a source of groundwater contamination.

Aeration, through enhanced volatilization of the contaminants in the vadose zone, removes the source of the contamination. Leachate will no longer be produced.

The RCRA cap reduces the amount of surface water percolating through the contaminated vadose zone soil and the leachate production. Untreated contaminated soils will remain on the site, however, posing a long term threat to groundwater quality. The cap maintenance required also reduces the permanence of this alternative.

The excavation and incineration alternative effectively and permanently treats the contaminated soil on the site. Disposal of the ash off-site represents a future environmental hazard and adds to the problem of hazardous waste disposal.

The pump and treat alternatives provide for containment and treatment

of the on-site contaminant groundwater and the off-site plume, treating to levels which permanently reduce the potential for further degradation of the aquifer. The pump and treat alternatives also eliminate the health risks of consuming contaminated groundwater and reduce the threat to downgradient municipal wells. Discharge concentrations levels from the treatment methods comply with NPDES permit standards. When clean-up levels have been reached, this alternative will provide both long term effectiveness and permanence.

4. **Reduction of Toxicity, Mobility or Volume** is the anticipated performance of the treatment technologies a remedy may employ. Toxicity of contaminants in the on-site soil will be reduced through treatment. The soil flushing alternative reduces the potential for off-site movement of the contaminants from soil by treatment of the flushing fluids. Groundwater treatment through pump and treat alternatives reduces the contamination of groundwater under the site and controls migration of the off-site plume.

5. **Short-term Effectiveness** involves the period of time needed to achieve protection as well as the adverse effect on human health and the environment that may be posed during the construction and implementation period until cleanup goals are achieved.

Short-term effects of the in-situ soil source control alternatives would be negligible. Contaminants are immobilized in the subsurface area and carried to the surface for treatment by a closed circuit extraction system which minimizes the potential for risk to the public health and the environment. Treatment of the VOCs in the aeration alternative and contaminant-laden fluid in the flushing alternative, operate in compliance with appropriate permit requirements.

The RCRA cap alternative introduces minor increases in the potential for exposure to subsurface soils and increases in the risk to the environment during the excavation period. The excavation and incineration alternatives could increase the risk of exposure to low levels of VOCs through either vapor emissions or particulate dispersion.

On-site groundwater source control and plume migration management alternatives depend upon groundwater usage restriction, in the form of advisories, for short term effectiveness. Emission treatment on the discharge side of the stripping tower will minimize exposure due to on-site treatment. The design of this system will meet the technical requirements of applicable air emission control regulations.

Short term effectiveness of the on-site groundwater treatment alternatives assumes that discharges of treated water from the larger capacity stripping system required to treat the on-site and off-site groundwater and the soil flushing fluids, will remain within the limits of the NPDES permit in effect for the currently operating system.

6. **Implementability** is the technical and administrative feasibility of a remedy, including the availability of goods and services needed to

implement the chosen solution.

A purge and treat system for the groundwater is a necessary component of the in-situ treatment of soil contamination by soil flushing. This system may be difficult to design due to the stratigraphic variability at the site. A pilot study may be required to determine the number of wells, air flow rates, and other parameters.

While capping or excavation and incineration are available technologies for leachate control, a limiting factor is the availability of a landfill and incinerator.

On-site groundwater control and on and off-site management of migration alternatives, properly designed, will provide containment, collection, and treatment, and meet the objectives of site remediation.

Air stripping technology can be designed to improve removal efficiency to meet POTW pretreatment standard and NPDES permit requirements. Discharge alternatives require use of existing lines to surface drainage or the construction of a pipeline to the existing sewer line.

7. Cost includes capital, operation and maintenance costs.

Total costs for the alternatives other than no action are compared in Table 9 below.

TABLE 9

<u>ALTERNATIVES</u>	<u>TOTAL COST</u>
On-site Source Control of Soils	
In-situ aeration	\$ 515,000
In-situ flushing	163,000
RCRA cap	1,003,000
Excavation and incineration	21,105,000
On-site Source Control of Groundwater with On-site Treatment and Off-site Disposal	
Air stripping	
surface discharge	\$ 2,369,500
POTW	5,237,700
Steam stripping	
surface discharge	5,313,500
POTW	8,171,700
Carbon adsorption	
surface discharge	3,180,100
POTW	6,048,300
* Relevant only if there is a No Action Alternative for off-site groundwater treatment.	

Management of Plume Migration with Off-site and On-site Collection and On-site Treatment with Off-site Disposal

 Air stripping

surface discharge	
10 ⁻⁶ plume	2,888,100
primary plume	3,359,500
FOIW	
10 ⁻⁶ plume	7,348,900
primary plume	8,651,900
Steam stripping	
surface discharge	
10 ⁻⁶ plume	8,321,600
primary plume	10,390,300
FOIW	
10 ⁻⁶ plume	12,772,400
primary plume	15,672,700
Carbon adsorption	
surface discharge	
10 ⁻⁶ plume	5,714,300
primary plume	6,023,200
FOIW	
10 ⁻⁶ plume	10,175,300
primary plume	11,315,600

8. Support Agency Acceptance indicates whether, based on its review of the RI/FS, the support agency (State) concurs, opposes, or has no comment on the preferred alternative.

The State of Michigan has been notified of USEPA's choice of a preferred alternative and concurs with the selected alternative.

9. Community Acceptance

A Responsiveness Summary detailing USEPA's response to each significant comment submitted during the comment period is attached to the ROD.

VIII. The Selected Remedy

The selected remedy for the U.S. Aviox site, soil flushing (Alternative A.1.b.) and collection of on-site and off-site groundwater with treatment by air stripping (Alternative C.1.), addresses both sources of contamination and the resulting off-site plume.

Control of the sources of contamination will be accomplished by flushing the contaminants out of the vadose zone soil. These flushing fluids will then be collected along with the on-site contaminated groundwater through a system of purge wells. Control and collection of off-site contaminated groundwater will be accomplished by a series of purge wells installed in the contaminant plume. The collection system will be designed to contain the plume and prevent further degradation of the aquifer.

The design phase will include studies of sufficient scope to determine operating parameters and flushing system efficiencies. The

effectiveness of the flushing system will be evaluated after it has been in operation for five years. This evaluation will be based on an analysis of soil and groundwater samples for indicator chemicals. If there is no significant reduction in the contaminant levels in soils and if, after regular sampling and analysis, we find that the groundwater contamination continues to exceed 10^{-6} total risk, an alternate source control remedy will be identified in a new Feasibility Study and justified in a new Record of Decision.

The proposed area of attainment for treatment of the contaminated groundwater is the 10^{-6} total risk plume. This 10^{-6} total risk plume is that volume of water which contains contaminants at concentrations that pose an increased lifetime risk of one cancer case in a population of one million people. Restoration of the plume will meet drinking water standards, eliminating the risk to public health through consumption of contaminated groundwater. Impact, if any, on the Niles municipal wells approximately two miles west of the site will also be reduced.

Determination of the potential impact on the Niles municipal wells in the USEPA Endangerment Assessment was based on a worst case scenario - no action with continuous release of contaminants at the site. Using this scenario, analysis indicated that no contaminants of concern would be at unacceptable levels at the municipal wells at the time of arrival of the plume after approximately seventy years. The proposed alternative treats the sources of potential contamination and virtually eliminates any potential for impact on the municipal wells.

The contaminated groundwater collected from the on-site and off-site purge wells will be treated on-site by air stripping. The treated groundwater will be discharged to the surface waters through local drainage in compliance with NPDES permit requirements. The efficiency of this treatment alternative will produce an effluent with contaminant levels in an order of magnitude less than the chronic toxicity criteria. Appropriate monitoring of the effluent will insure that permit requirements are met during operation. Installation of purge wells and collection or discharge systems for off-site groundwater will meet state and local regulations regarding easements and construction standards. Discharge of VOCs to the atmosphere will be managed through emission controls designed to meet the requirements of state construction and operation air quality permits. Monitoring will insure that air quality standards are met.

There will be no additional risks to the community or on-site workers during the flushing, collection and treatment of the contaminated groundwater by the air stripper. Workers will use appropriate protective clothing (including respiratory protection) and adhere to an approved site safety plan.

The total estimated cost for the proposed remedies is \$3,024,100. This estimated cost includes the on-site flushing of the contaminated soil (\$163,000) and the cost of the treatment of on-site and off-site contaminated groundwater (\$2,888,100).

IX. Statutory Determinations

Section 121 of CERCLA requires that, at a minimum, the selected remedy shall:

- * Be protective of human health and the environment.
- * Attain ARARs or provide grounds for invoking a waiver.
- * Be cost-effective.
- * Utilize permanent solutions and alternative treatment technologies or resource recovery technologies to the maximum extent practicable.
- * Address whether the preference for a treatment that reduces toxicity, mobility, or volume as a principle element is satisfied, or provides an explanation as to why it is not satisfied.

A. Protection of Human Health and the Environment

The existing and potential risk to human health come about through exposure to contaminated groundwater and from the off-site plume through consumption and domestic use. This risk is limited to accidental or uninformed use only, since a municipal water system has been extended into the affected area. On a long term basis, there is the potential for exposure beyond the current area of contamination if the plume continues to migrate. The selected remedy eliminates potential risks and ensures adequate protection of human health and the environment through containment and treatment of the on-site plume within the area of attainment to 10^{-6} risk levels. This level of treatment represents contaminant levels which, at a minimum meet drinking water standards.

Short term risks may be posed during the treatment process by the discharge of treated water to surface drainage and by emissions to the atmosphere from the air stripper. However, the treatment process will, treat groundwater to levels which will meet the NPDES permit requirements and cause no unacceptable short term risks by implementation of the remedy. Atmospheric emissions will meet the standards of the Federal and State Air Pollution Control Acts.

B. Attainment of Applicable or Relevant and Appropriate Requirements

The applicable or relevant and appropriate Federal and State requirements (ARARs) that were considered for the U.S. Aviox site and satisfied through the selected remedy are listed in Table 10. The selected remedy attains Federal and State ARARs.

Chemical specific ARARs address the clean up levels for indicator

TABLE 10

U.S. Aviox
Selected Remedy ARARs

I. Federal Requirements

- | | |
|--|--|
| A. Groundwater Protection | |
| 1. Clean up levels for indicator chemicals (Benzene, TCE, 1,2-DCA, 1,1-DCE, 1,1,1-TCA) | 40 CFR 141 |
| B. Surface Water Protection | |
| 1. Regional management program | CWA, Section 208 |
| 2. NPDES - discharge of treated groundwater | 40 CFR 122
40 CFR 125
40 CFR 131
40 CFR 236 |
| C. Construction of Treatment System and Disposal of Treatment Residuals | 40 CFR 260-263 |
| D. Protection of air quality during Treatment | Clean Air Act,
Section 101
40 CFR 50, 52 |
| E. Generation, Transportation, Storage, Treatment and Disposal of Hazardous Wastes | RCRA Subtitle C
40 CFR 260-264 |

II. State of Michigan Requirements

- | | |
|--|---|
| A. Surface Water Protection | |
| 1. Discharges from groundwater treatment | Act 245, Part 9, 21 |
| B. Air Quality Protection | Act 348 |
| C. Treatment System Installation | Act 98
Act 846, Section 3
Act 315 |
| D. Treatment Residuals | Act 64, Parts 2, 3,
5, 6 |
| E. Hazardous Wastes | Act 64 |

chemicals. The Safe Drinking Water Act, 40 CFR 141, establishes maximum contaminant levels (MCLs) for drinking water quality. MCLs will be reached or exceeded within the 10^{-6} risk level area of attainment. Action specific ARARs relate to the installation and operation of the groundwater collection and treatment alternative, the discharge of treated groundwater, air emissions, and disposal of treatment generated wastes.

Permit requirements for installation of wells in the groundwater collection system pursuant to Michigan Act 315, are relevant and appropriate and will be met. RCRA Subtitle C, 40 CFR 260 through 264 and 40 CFR 100 through 199, regulate the generation, transportation, storage, treatment, and disposal of hazardous wastes generated in the course of remedial action. The construction design, monitoring operation, and closure of hazardous facilities is also regulated. This program is administered by MDNR under Act 64. The requirements are applicable to the groundwater system and to the disposal of treatment residuals. These requirements will be met.

Discharge of treated groundwater to surface waters is regulated by the Clean Water Act Section 402, 40 CFR 122 and 125, Subchapter N, National Pollution Discharge Elimination System (NPDES) which regulates point discharges to navigable waters. The Clean Water Act is administered by MDNR under Michigan Act 245. NPDES requirements apply to discharges to the Bame-Huntly drain. Treated waters will be discharged at levels which meet permit requirements.

The Clean Air Act, 40 CFR 50 and 52, implements and sets rules for the regional air pollution control program and establishes Ambient Air Quality Standards. The program is administered by the MDNR under Michigan Act 348. These requirements are applicable to air emissions generated during treatment of groundwater. Emissions during operation of the air stripper will be monitored and controlled to comply with Federal PSD regulations, Federal NSPS and NESHAP standards, and permitting requirements.

C. Cost Effectiveness

Total cost for the selected remedy is \$3,024,100. The soil flushing alternative for treatment of contaminated soil is \$163,000 and the treatment of on-site and off-site contaminated groundwater is \$2,888,100. This remedy provides the best overall effectiveness commensurate with costs such that it represents the most reasonable value for the money.

A discussion comparing other alternatives considered for treatment of on-site soil contamination follows below:

Aeration	\$	515,000
RCRA Cap		1,003,000
Excavation and Incineration		21,105,000

Costs for the RCRA cap are one order of magnitude greater than the

selected remedy and offer less protectiveness. Incineration is two orders of magnitude greater than the preferred remedy and offers no increase in protectiveness. The cost differential between aeration and soil flushing is a factor of two with no increase in protectiveness for the aeration alternative.

.. Costs for collection of groundwater for on-site and off-site control of migration are the same for each of the treatment alternatives. Costs vary according to the treatment applied, the method of disposal, and the volume of treated groundwater. No advantage in protectiveness results from disposal at the POTW over discharge to surface waters. Both methods of disposal meet NPDES permit requirements for discharge to surface waters. The groundwater is treated to meet 10^{-6} risk levels providing protectiveness to meet or exceed drinking water standards. Treatment costs for contaminated groundwater are compared below:

Air stripping	\$ 2,888,100
Steam stripping	8,321,600
Carbon adsorption	5,714,300

D. Utilize Permanent Solutions and Alternative Technologies

The selected remedy provides a permanent solution employing alternative treatment or resource recovery technologies to the maximum extent practicable and provides the best balance among the nine evaluation criteria of all the alternatives examined. The remedy is the most appropriate solution for the site. The in-situ soil treatment alternative satisfies the preference for a permanent solution, alternative treatment technology, and attains ARARs. The soil flushing alternative demonstrates an advantage over the aeration alternative because of site conditions which may reduce the effectiveness of the aeration alternative.

Design efficiencies for each of the treatment alternatives will meet surface discharge permit requirements. Air stripping affords overall effectiveness equal to the steam stripping and carbon adsorption alternatives and is the most cost effective.

Groundwater treatment alternatives also satisfy the preference for permanent solutions and alternative technology and attain ARARs. The air stripping alternative is the most cost effective. The remedy satisfies the preference for treatment that addresses the principal threats posed by the site.

E. Reduction of Toxicity, Mobility or Volume

Toxicity of contaminants in the on-site soil will be reduced through treatment. The soil flushing alternative reduces the potential for off-site movement of the contaminants from soil by treating the soil and the flushing fluids. Groundwater treatment through pump and treat alternatives reduces the contamination of groundwater under the site and controls migration of the off-site plume.

Item 9

REGION : 5
LOCATION : NILES MI
SITE NAME : U067113
RCD ID : R03-BB7023
SITE ID : MID980794556
RCD DATE : 890907
CONTAMINANTS : VOCs

BENZENE
TCA
TCE
TOLUENE
XYLENES

REMEDY :

THE SELECTED REMEDY TREATS THE ON-SITE SOIL CONTAMINATED WITH VOLATILE ORGANIC COMPOUNDS (VOCs) BY FLUSHING. GROUNDWATER, BOTH ON-SITE AND OFF-SITE, CONTAMINATED BY VOCs AND FLUIDS FROM THE SOIL FLUSHING ALTERNATIVE ARE COLLECTED AND TREATED ON-SITE BY AIR STRIPPING. A PUBLIC WATER SUPPLY HAS BEEN PROVIDED TO THE AFFECTED AREA UNDER A SEPARATE ACTION.

ABSTRACT :

THE SIX-ACRE U.S. AVIEX SITE IS LOCATED IN THE CITY OF NILES, HOWARD TOWNSHIP, CASS COUNTY, MICHIGAN. THE SITE LIES IN A RESIDENTIAL NEIGHBORHOOD WITH SINGLE FAMILY HOMES 100 FEET TO THE SOUTH, EAST, AND NORTH. BECAUSE THERE ARE NO STORM OR SANITARY SEWERS NEARBY, SURFACE RUNOFF FROM THE SITE AND SURROUNDING AREAS FLOWS TOWARD BOME-HUNTLY DRAIN, A TRIBUTARY OF ST. JOSEPH RIVER, AND ULTIMATELY LAKE MICHIGAN. U.S. AVIEX PRODUCED NON-LUBRICATING AUTOMOTIVE FLUIDS FROM THE EARLY 1960S UNTIL 1978. DURING OPERATIONS AT THE PLANT, CHLORINATED HYDROCARBONS, INCLUDING TRICHLOROETHANE (TCA), WERE RELEASED INTO THE VADOSE ZONE SOUTH OF THE PROCESS ROOM. THE SUBSURFACE SOIL STILL CONTAINS SIGNIFICANT QUANTITIES OF TCA, TCE, AND PCE. IN JULY 1972, AN UNDERGROUND PIPELINE CONTAINING DIETHYL ETHER WAS BROKEN

C:\HAYES\FLUSH

Thursday August 17, 1989 10:14:00 am

DURING EXCAVATION NEAR THE WEST WALL OF THE PACKAGING ROOM. OVER THE NEXT SEVERAL YEARS, DIETHYL ETHER CONTAMINATION WAS IDENTIFIED IN DOWNGRADE RESIDENTIAL WELLS SOUTHWEST OF THE PLANT, APPROXIMATELY 0.15 MILE AWAY. U.S. AVIEX INSTALLED SIX MONTH MONITORING WELLS ON THE SITE BEGINNING IN 1972 AND ALSO PROVIDED ALTERNATE WATER SUPPLIES TO HOMES WITH CONTAMINATED WELLS. IN NOVEMBER 1978, A FIRE OCCURRED ON THE SITE, AND THE LARGE VOLUMES OF WATER USED TO EXTINGUISH THE FIRE FLUSHED UNKNOWN AMOUNTS OF CHEMICALS CONTAINED IN BARRELS AND TANKS INTO THE SOIL. FOLLOWING THE FIRE, CHLORINATED HYDROCARBONS WERE DETECTED IN DOMESTIC WELLS. BASED ON DATA DEVELOPED DURING A GROUND WATER INVESTIGATION IN 1982, A PURGE AND TREATMENT SYSTEM USING AIR STRIPPING AND DISCHARGE TO SURFACE WATER WAS DESIGNED AND INSTALLED TO CONTAIN THE CONTAMINATION. CONTAMINATED GROUND WATER, HOWEVER, IS CURRENTLY FLOWING OFFSITE TO THE SOUTHWEST INTO RESIDENTIAL AREAS. THE PRIMARY CONTAMINANTS OF CONCERN AFFECTING SOIL AND GROUND WATER ARE VOCs INCLUDING BENZENE, PCE, TCE, TOLUENE, AND XYLENES.

THE SELECTED REMEDIAL ACTION FOR THIS SITE INCLUDES: SOIL FLUSHING OF APPROXIMATELY 11,500 YDS OF CONTAMINATED ON-SITE SOIL; AND COLLECTION OF ON-SITE AND OFF-SITE GROUND WATER AND FLUIDS FROM THE SOIL FLUSHING PROCESS WITH TREATMENT ON-SITE BY AIR STRIPPING, AND DISCHARGE TO SURFACE WATER. THE ESTIMATED PRESENT WORK COST FOR THIS REMEDIAL ACTION IS \$3,024,000.

Exhibit E

Item 2

REGION : 2
LOCATION : PLUMSTED, NJ
SITE NAME : GOOSE FARM
ROD ID : R02-85/016
SITE ID : NJD980030109
ROD DATE : 850927
CONTAMINANTS : ETHYLBENZENE
PCBS
TOLUENE
TRICHLOROETHYLENE (TCE)
VOCs

REMEDY :

1. FLUSH THE CONTAMINATED SOIL AND GROUNDWATER UNDERLYING THE SITE. THE GROUNDWATER WOULD BE RECOVERED USING A WELLPOINT SYSTEM AND TREATED ON-SITE PRIOR TO REINJECTION INTO THE SOIL. CURRENTLY, IT IS ESTIMATED THAT RECOVERING AND FLUSHING TEN PORE VOLUMES WILL BE REQUIRED TO REMOVE THE MOBILE CONTAMINANTS FROM THE SOIL AND GROUNDWATER. PILOT STUDIES TO BE CONDUCTED DURING DESIGN WILL OPTIMIZE THE REQUIRED TREATMENT SYSTEM COMPONENTS.
2. FOLLOWING SOIL FLUSHING AND GROUNDWATER RECOVERY AND TREATMENT, CONDUCT AN EXTENSIVE TESTING PROGRAM TO DETERMINE THE NEED TO CAP THE SITE.
3. DURING AND AFTER SOIL FLUSHING AND GROUNDWATER RECOVERY AND TREATMENT ACTIVITIES, CONDUCT AN EXTENSIVE TESTING PROGRAM TO DETERMINE THE EXTENT OF PCB CONTAMINATION IN THE FORMER DRUM PIT AREA BASED ON THIS PROGRAM, DETERMINE THE NEED, IF ANY, TO REMEDIATE SOIL.

ABSTRACT :

THE GOOSE FARM SITE IS LOCATED APPROXIMATELY TWO MILES NORTHEAST OF THE TOWN OF NEW EGYPT IN PLUMSTED TOWNSHIP, OCEAN COUNTY, NEW JERSEY. THE GOOSE FARM WAS USED AS A HAZARDOUS WASTE DISPOSAL SITE FROM THE MID 1940'S TO THE MID 1970'S BY A MANUFACTURER OF POLYSULFIDE RUBBER AND SOLID ROCKET FUEL PROPELLANT. THE MAJORITY OF WASTES WERE DUMPED INTO A PIT DUG THROUGH THE FINE SAND. THE DIMENSIONS OF THE PIT WERE APPROXIMATELY 100 X 300 X 15 FEET. LAB PACKS, 55 GALLON DRUMS, AND BULK LIQUIDS WERE DUMPED INTO THE PIT. INVESTIGATIONS HAVE FOUND CONTAMINATED SOILS CONTAINING VOLATILE, ACID AND BASE/NEUTRAL ORGANIC POLLUTANTS THROUGHOUT THE DISPOSAL AREA. IN ADDITION, SAMPLING SHOWS CONTAMINATION OF GROUND WATER UP TO 570 PPM TOTAL PRIORITY POLLUTANTS AND CONTAMINATION OF THE SURFACE WATER UP TO 1100 PPB TOTAL VOLATILE ORGANICS.

THE RECOMMENDED REMEDIAL ALTERNATIVE FOR THIS SITE IS EXPECTED TO BE IMPLEMENTED IN A PHASED MANNER. FIRST, THE CONTAMINATED SOIL AND GROUND WATER UNDERLYING THE SITE WILL BE FLUSHED. THE GROUND WATER WILL BE RECOVERED USING A WELL-POINT SYSTEM AND WILL BE TREATED ON SITE PRIOR TO REINJECTION INTO THE SOIL. FOLLOWING SOIL FLUSHING AND GROUND WATER RECOVERY AND TREATMENT, EXTENSIVE TESTING WILL BE CONDUCTED TO DETERMINE THE NEED TO CAP THE SITE. IN ADDITION, DURING AND AFTER SOIL FLUSHING AND GROUND WATER RECOVERY AND TREATMENT ACTIVITIES, EXTENSIVE TESTING WILL BE CONDUCTED TO DETERMINE THE EXTENT OF PCB CONTAMINATION IN THE FORMER DRUM PIT AREA. TEST DATA WILL DETERMINE THE NEED TO REMEDIATE PCB-CONTAMINATED SOIL. IF SUCH REMEDIATION IS DEEMED NECESSARY, A SUPPLEMENTARY RECORD OF DECISION WILL BE PREPARED. TOTAL CAPITAL COST FOR THE SELECTED REMEDIAL ALTERNATIVE IS ESTIMATED TO BE \$3,014,000 WITH NO O&M COSTS.